

# Integration of an SiPM-Based Scintillation Detector into the Phase-2 CMS Drift Tube Readout Chain Using the IpGBT ASIC

Master's thesis in physics

Nils Jonathan Esper

Supervisor: Univ.-Prof. Dr. rer. nat. Alexander Schmidt

Second reviewer: Apl. Prof. Dr. rer. nat. Oliver Pooth



Physics Institute III A, RWTH Aachen University

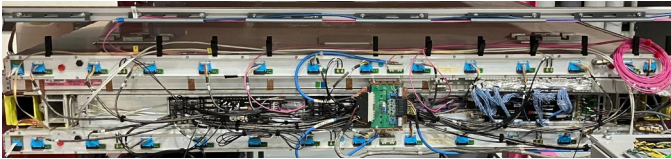
Aachen, 19<sup>th</sup> December 2025

# Introduction

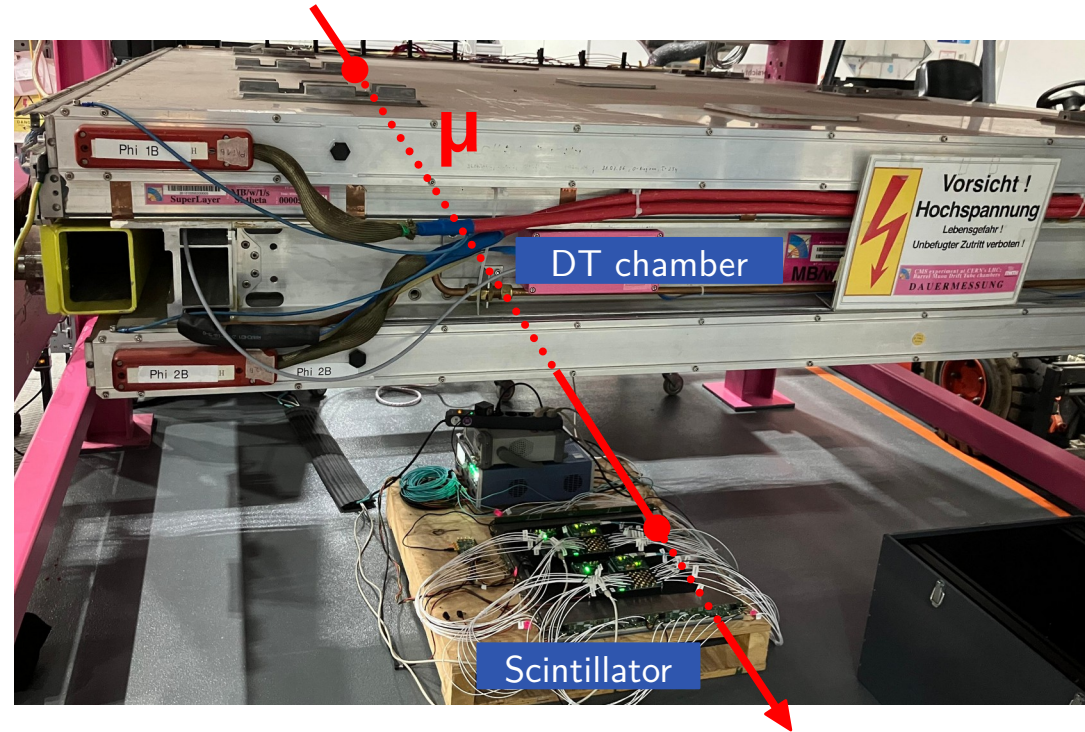
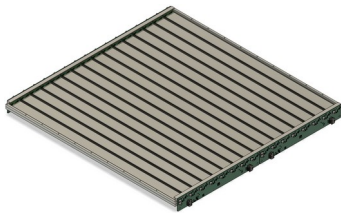
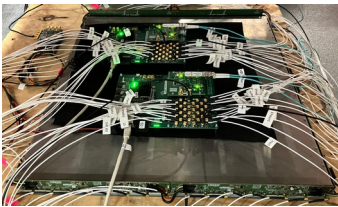
# Introduction

Two different muon detectors available:

- CMS Drift Tube (DT) chamber:



- Scintillator strip detector with SiPM readout:



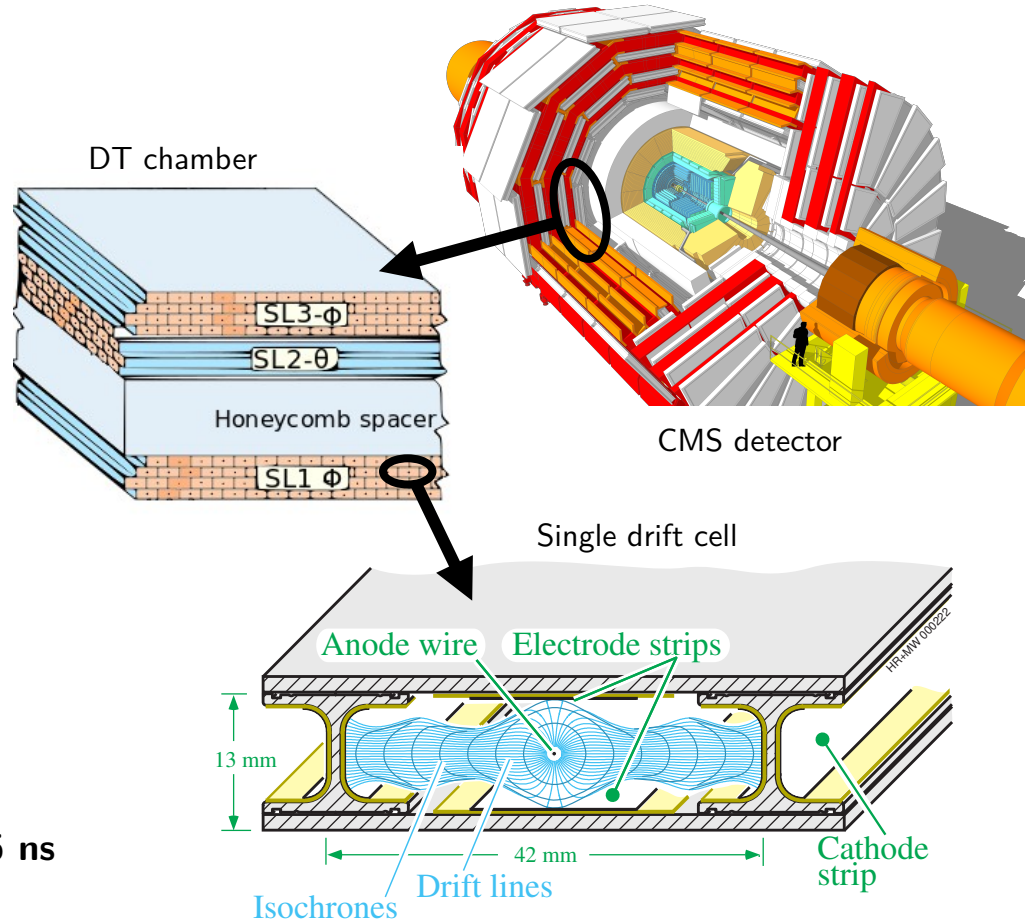
Idea for this thesis:

- **Integrate scintillator into the CMS DT readout chain at cosmic test stand**
  - **Demonstrate synchronization of detectors with atmospheric muon data**

# CMS Drift Tube chamber

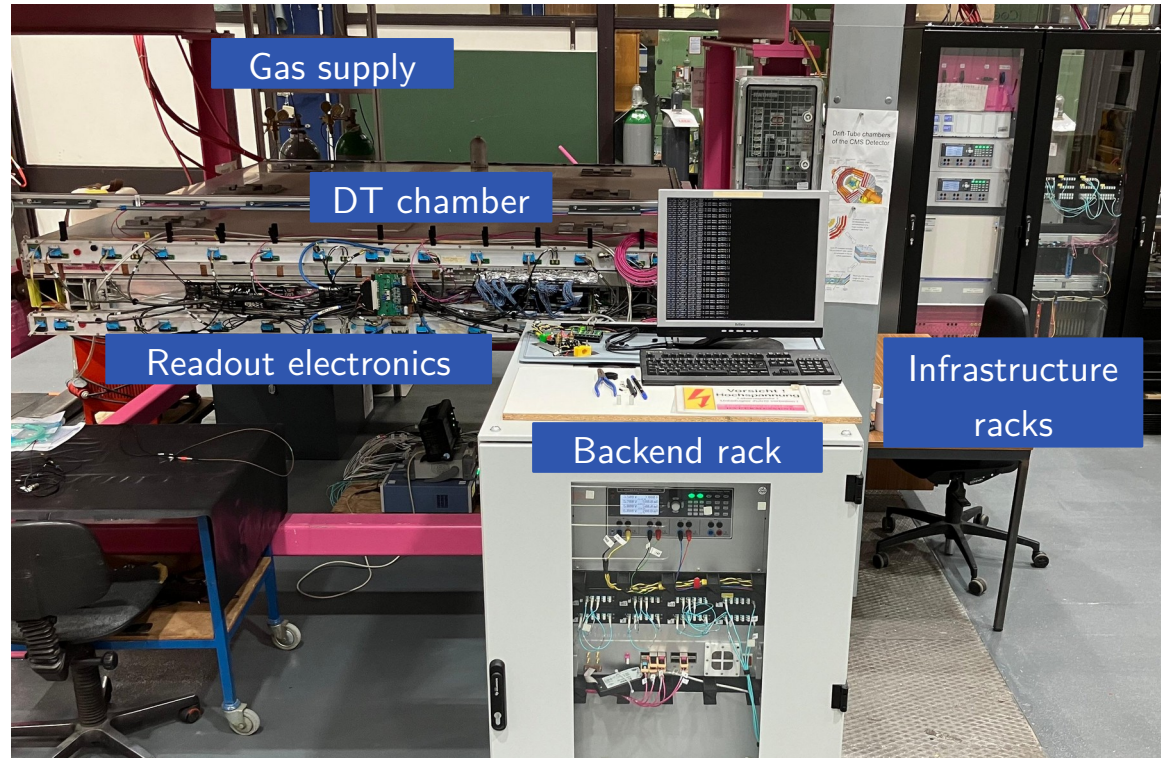
# CMS DT system

- Compact Muon Solenoid (CMS) detector:
  - Multipurpose particle detector at Large Hadron Collider (LHC) at CERN
- Drift Tube (DT) chambers:
  - Gaseous detector in CMS barrel region  $|\eta| < 1.2$
  - **3 superlayers** with **4 layers** of  $O(50)$  **drift cells** each
  - SL 2 rotated relative to SLs 1, 3
- Drift cell:
  - Cross section: **42 x 13 mm<sup>2</sup>**
  - Length:  $O(2 \text{ m})$
  - Gas mixture: 85% Ar, 15% CO<sub>2</sub>
  - Electric field:  $O(1.5 \text{ kV/cm})$
  - Ionization electrons drift with  $\mathbf{v_{drift} = 54.5 \text{ } \mu\text{m/ns}}$
  - Cell half-width determines max. drift time:  $\mathbf{t_{drift,max} = 385 \text{ ns}}$



# Cosmic test stand in Aachen

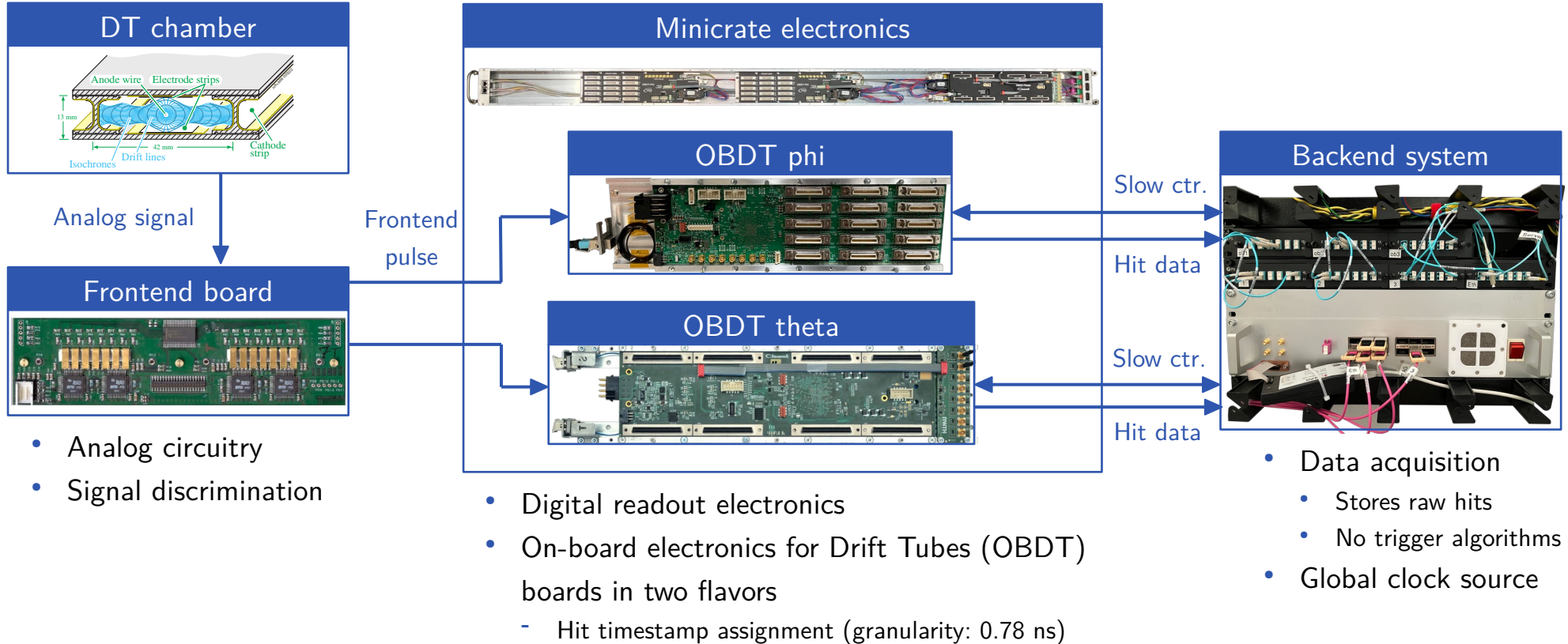
- Our institute operates cosmic test stand in experimental hall
  - 25% of DT chambers for CMS produced in Aachen in early 2000s
  - Spare chamber left at test stand
- Features fully functional CMS DT chamber
  - Dimensions:  $2.5 \times 2.2 \times 0.4 \text{ m}^3$



Setup at cosmic test stand

# DT readout electronics

- Resembles CMS DT Phase-2 readout chain



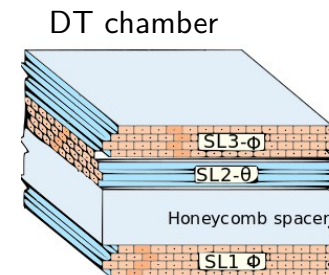
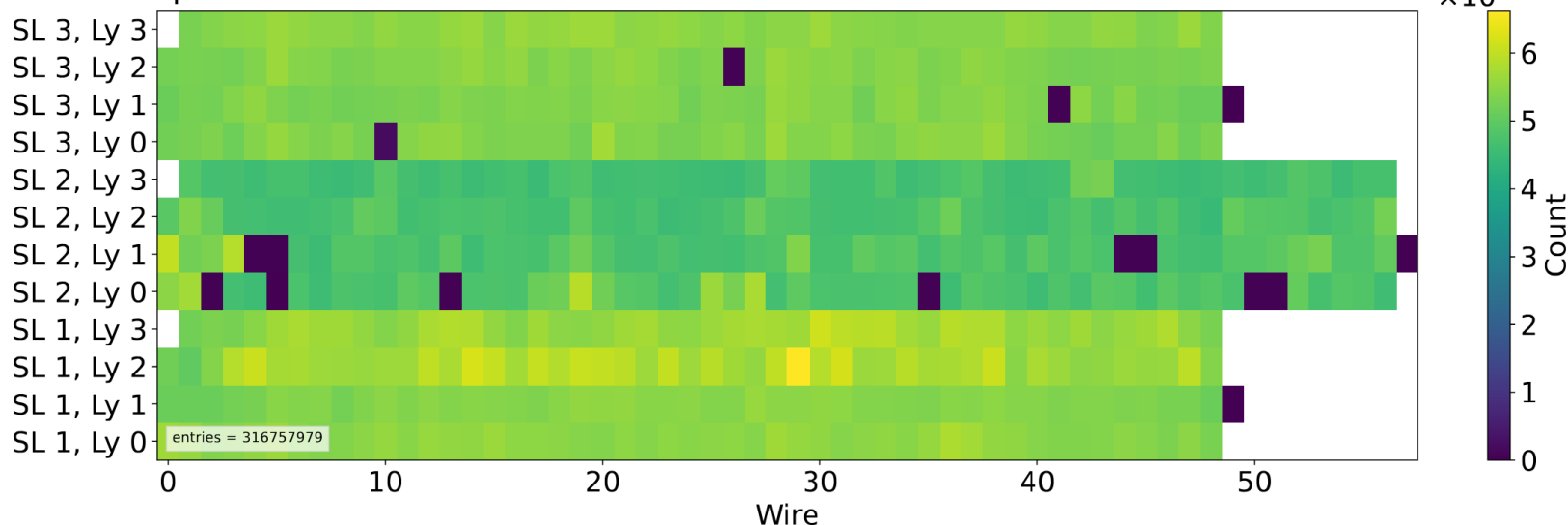
# Recorded Drift Tube chamber data

# DT hit data

- Recorded large dataset of atmospheric muons with DT chamber:

| Duration | DT hits     |
|----------|-------------|
| 5h 27min | 316,757,979 |

Wire occupancies:



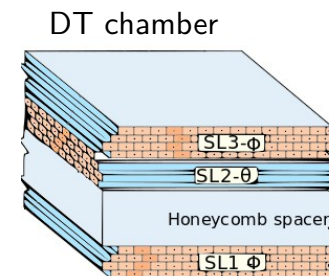
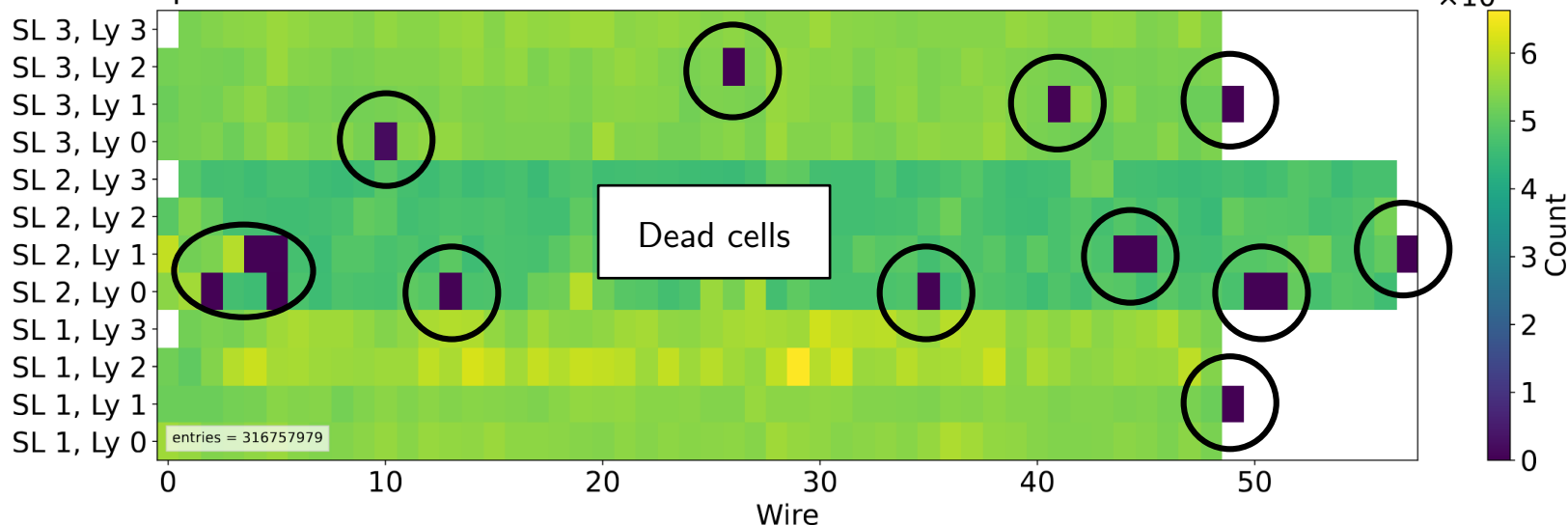
- Observe uniform wire rates: **28.0 Hz** (SLs 1, 3) and **24.5 Hz** (SL 2)
  - Rate differences because of different wire lengths (chamber size:  $2.5 \times 2.2 \text{ m}^2$ )
- 16 dead cells** in the chamber

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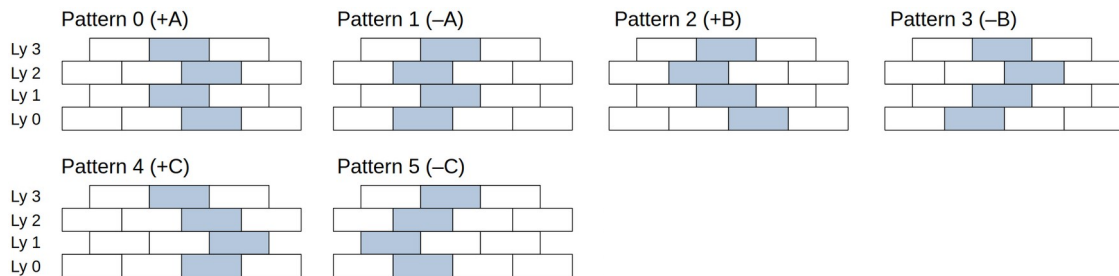
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# DT superlayer patterns

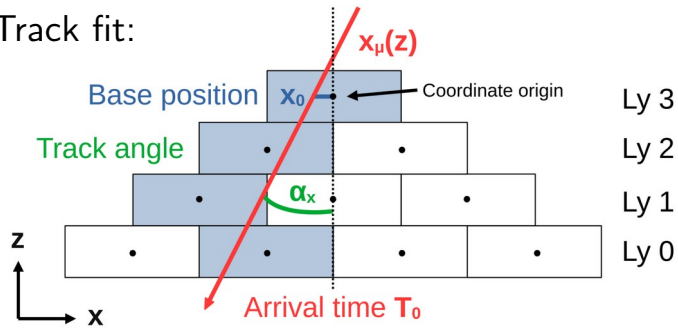
## Triggerless track reconstruction:

- ▶ Treat all cell hits as potential track hits
- Superlayer pattern:
  - Exclude hits that could not be created simultaneously:  $\Delta T_{\max} \leq t_{\text{drift},\max}$
  - Pattern should have **track-like shape**
  - Require 4 hits in pattern
- Superlayer pattern fit:
  - **Fit local track** to pattern hits
    - Relate hit timestamps to linear track
    - Assume horizontal drift with  $v_{\text{drift}} = \text{const.}$
    - **Local track parameters:**  $T_0$ ,  $x_0$ ,  $\tan \alpha$
  - **Apply cuts** to reduce noise patterns
    - Goodness-of-fit variable  $\chi^2/N_{\text{df}} < 10$
    - Remove unphysical drift times

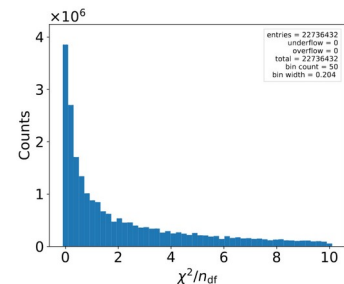
## Superlayer pattern geometry:



## Track fit:



## $\chi^2/N_{\text{df}}$ distribution:

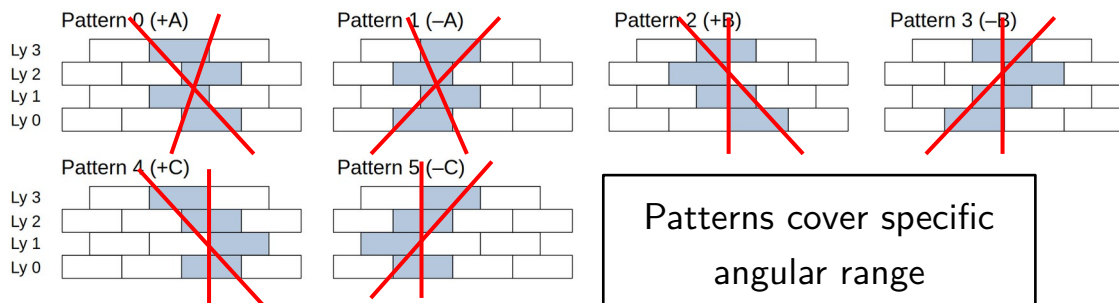


# DT superlayer patterns

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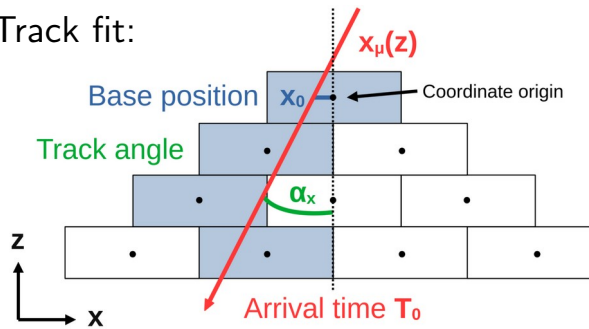
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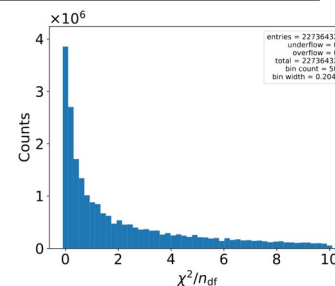


Patterns cover specific angular range  
→ Reconstruction only up to max. angle

## Track fit:

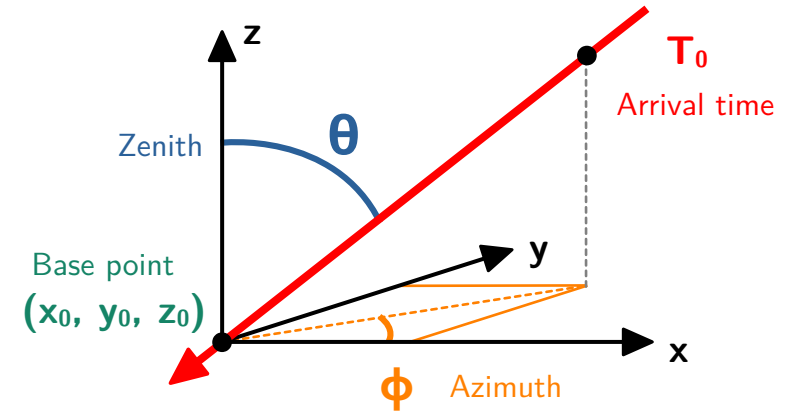


Ly 3  
Ly 2  
Ly 1  
Ly 0



# DT global track building

- Temporal matching of pattern fits:
  - Require fitted pattern in all 3 superlayers
  - Select patterns close in time:  $\Delta T_{0,SL} \leq 20 \text{ ns}$
- Translation of track projections into global frame:
  - **Global track parameters:**  $T_0$ ,  $(x_0, y_0, z_0)$ ,  $\phi$ ,  $\theta$



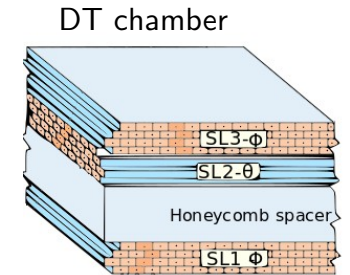
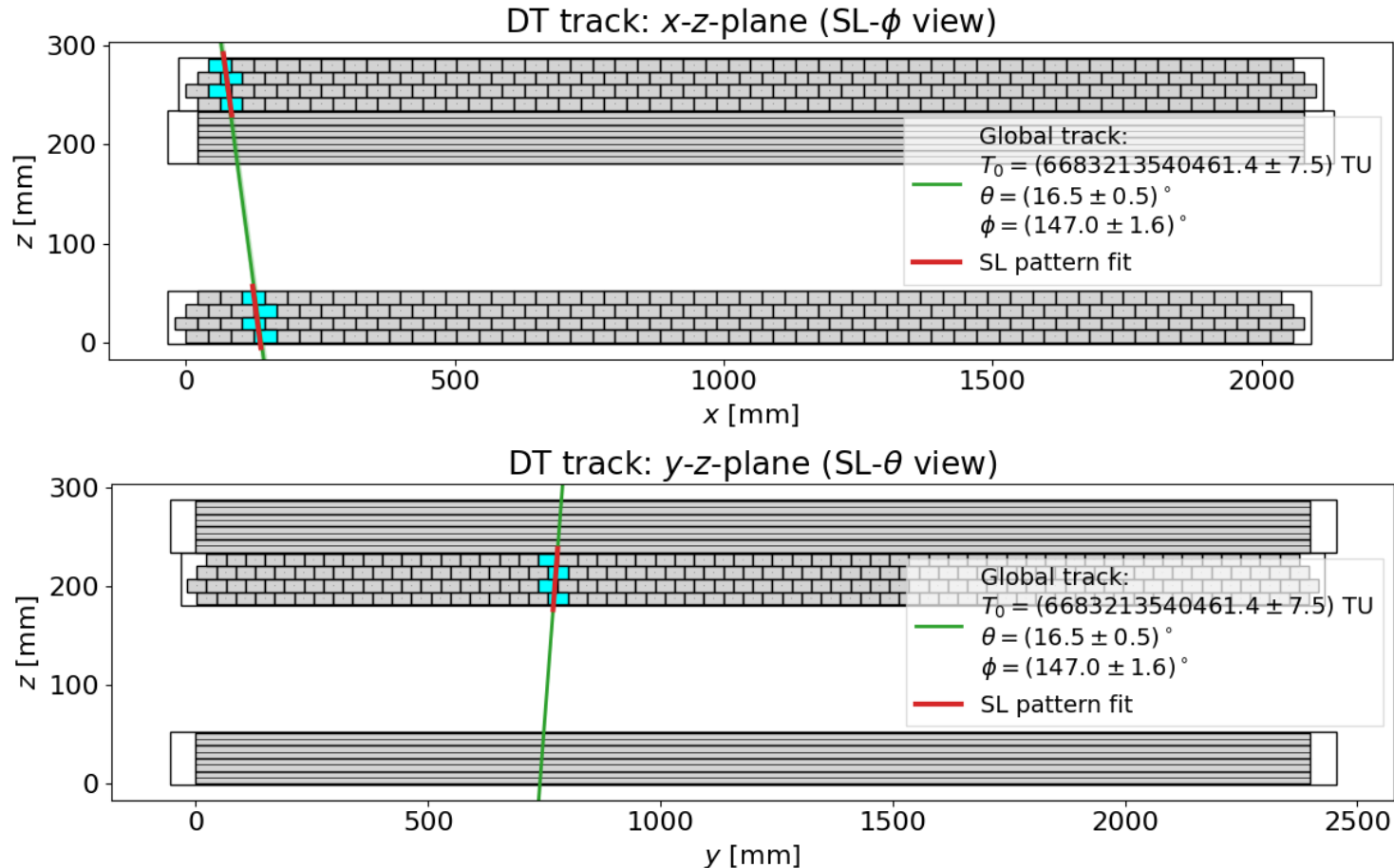
- ▶ Reconstructed global muon rate: **85 Hz**
  - ▶ Global track parameters (mostly) behave as expected

Simplified atmospheric muon simulation: **196 Hz**

- ▶ Should be understood as upper limit
  - ▶ Simulated hits are generated according to fit assumptions
  - ▶ No modeling of noise or (drift time) measurement uncertainties

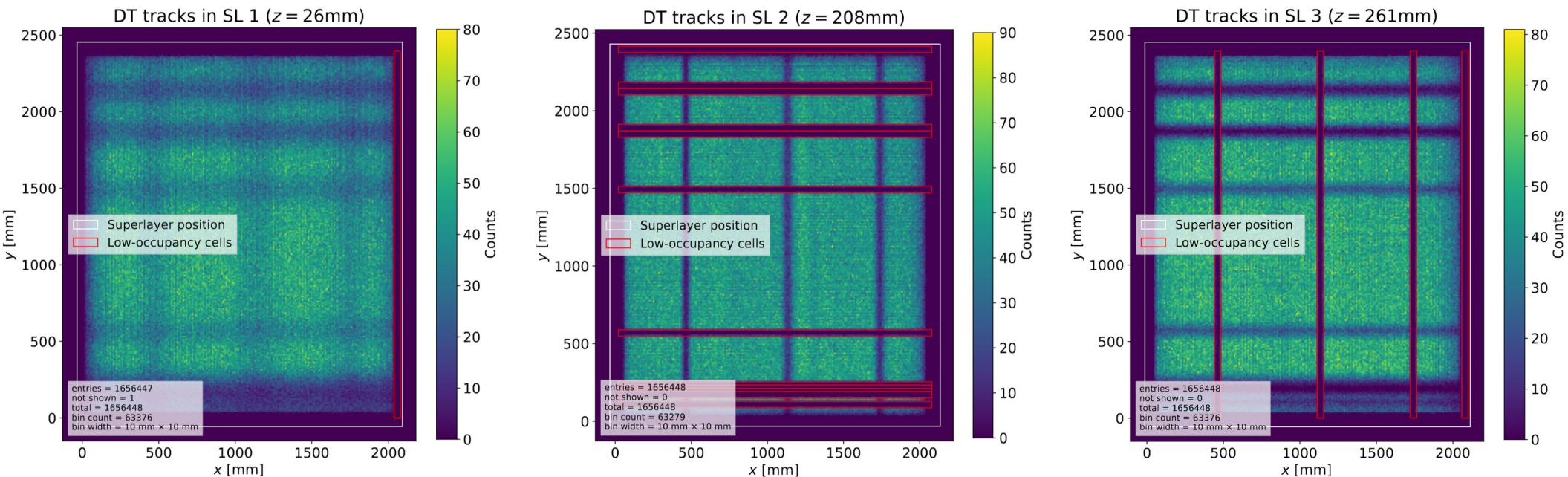
# DT global track results

Example reconstructed track, lateral view:



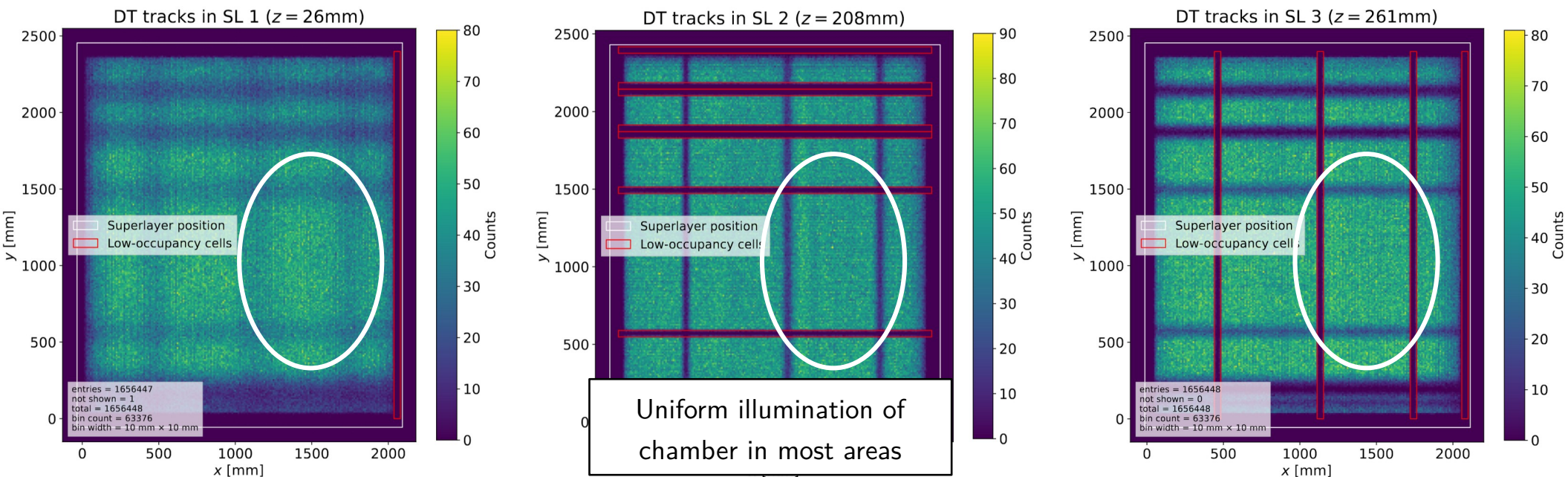
# DT global track results

2D histograms of global track positions in superlayers, top view:



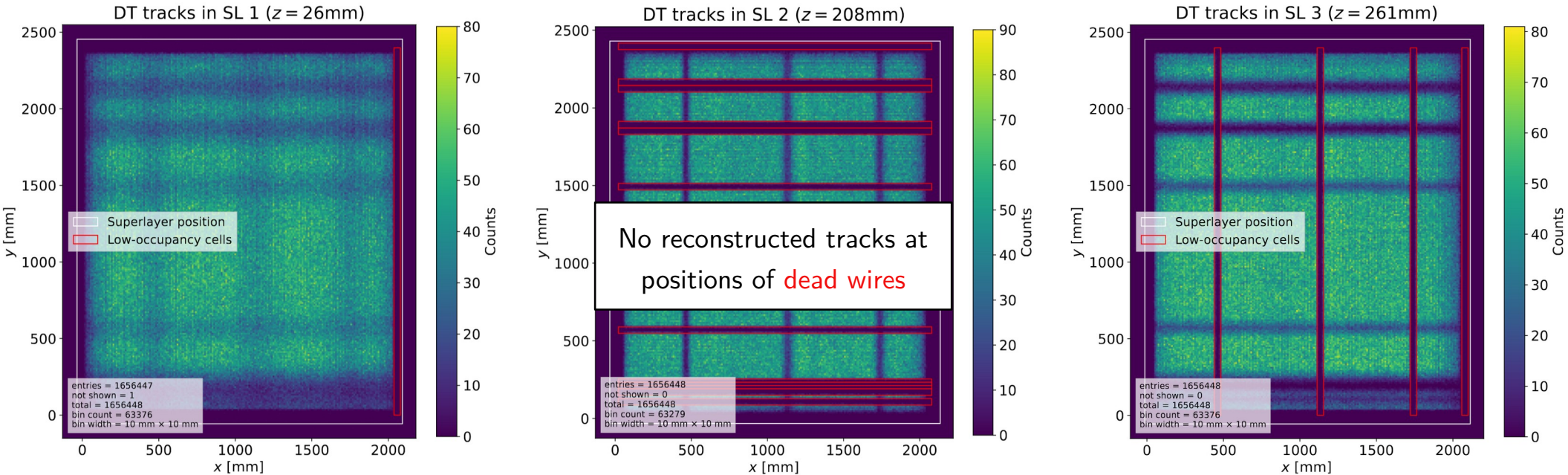
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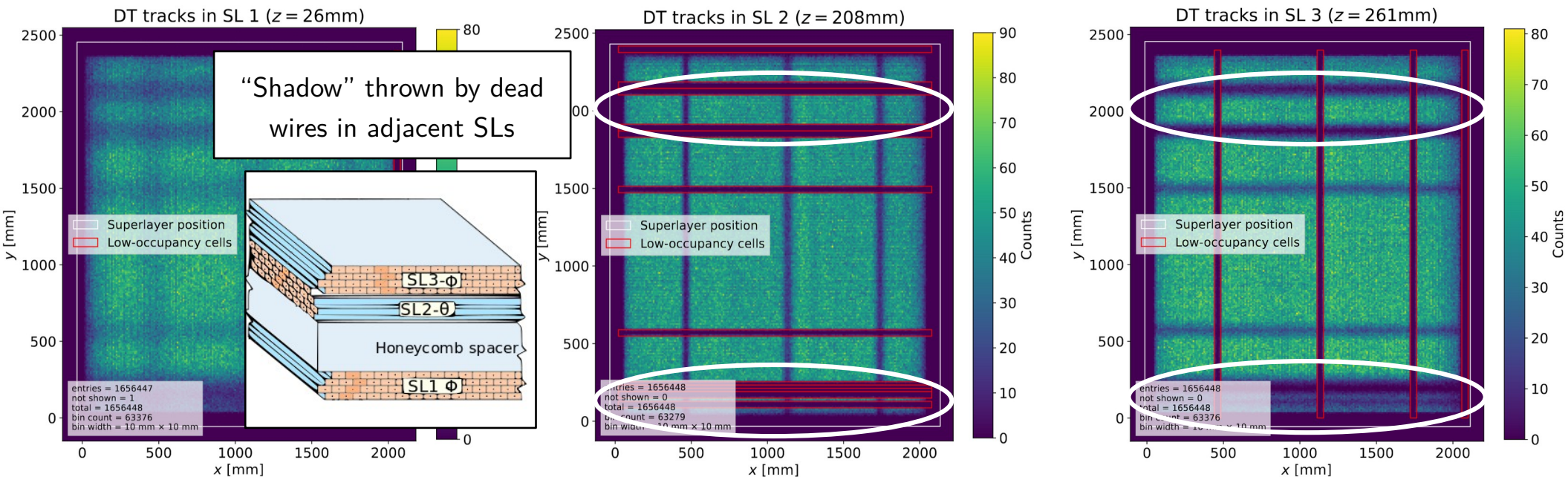
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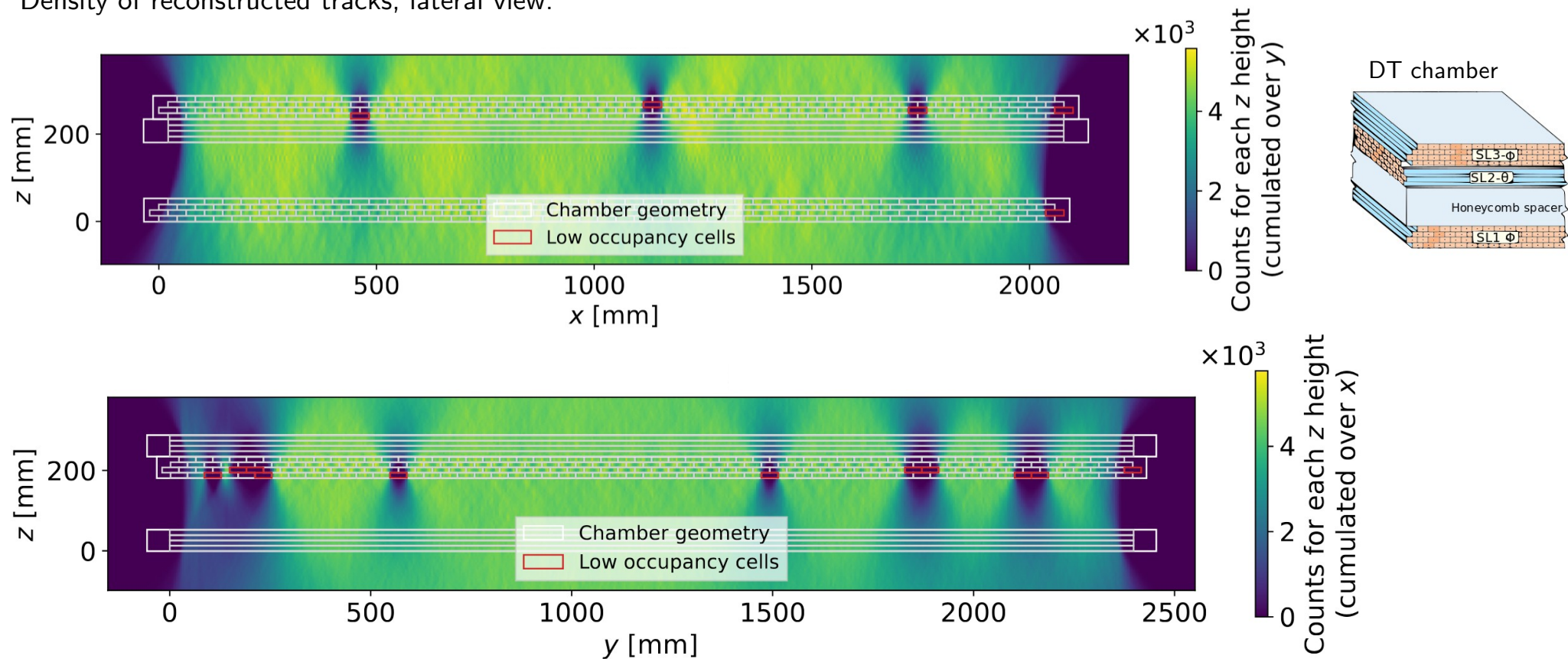
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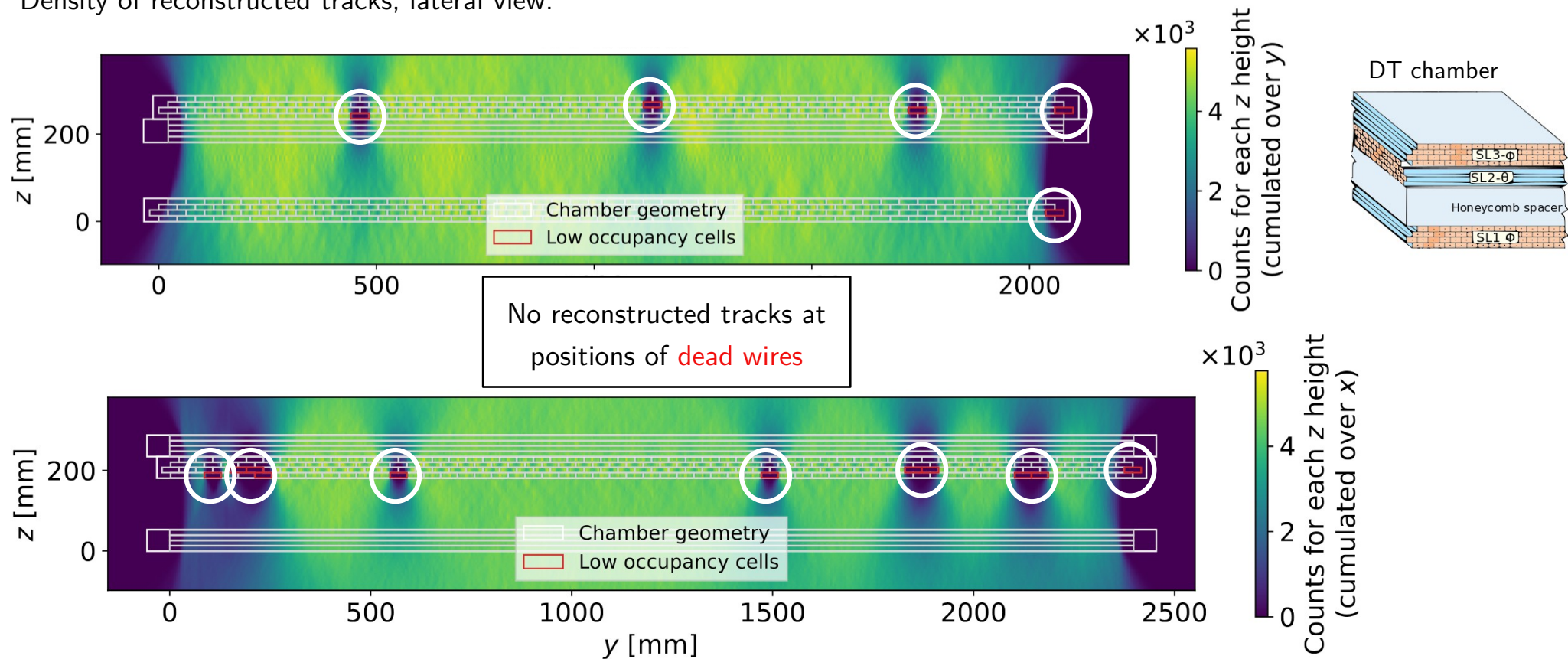
# DT global track results

Density of reconstructed tracks, lateral view:



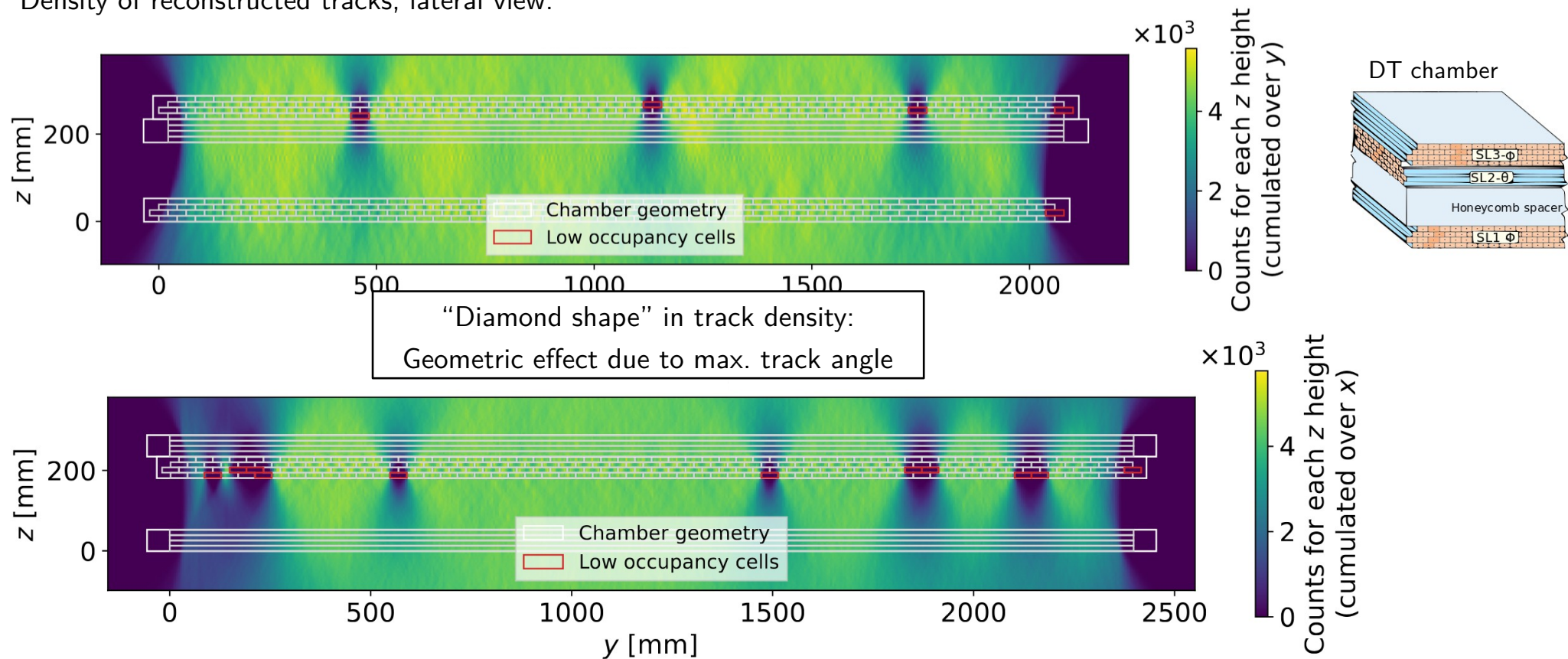
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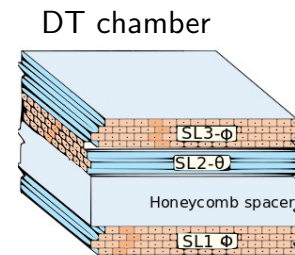
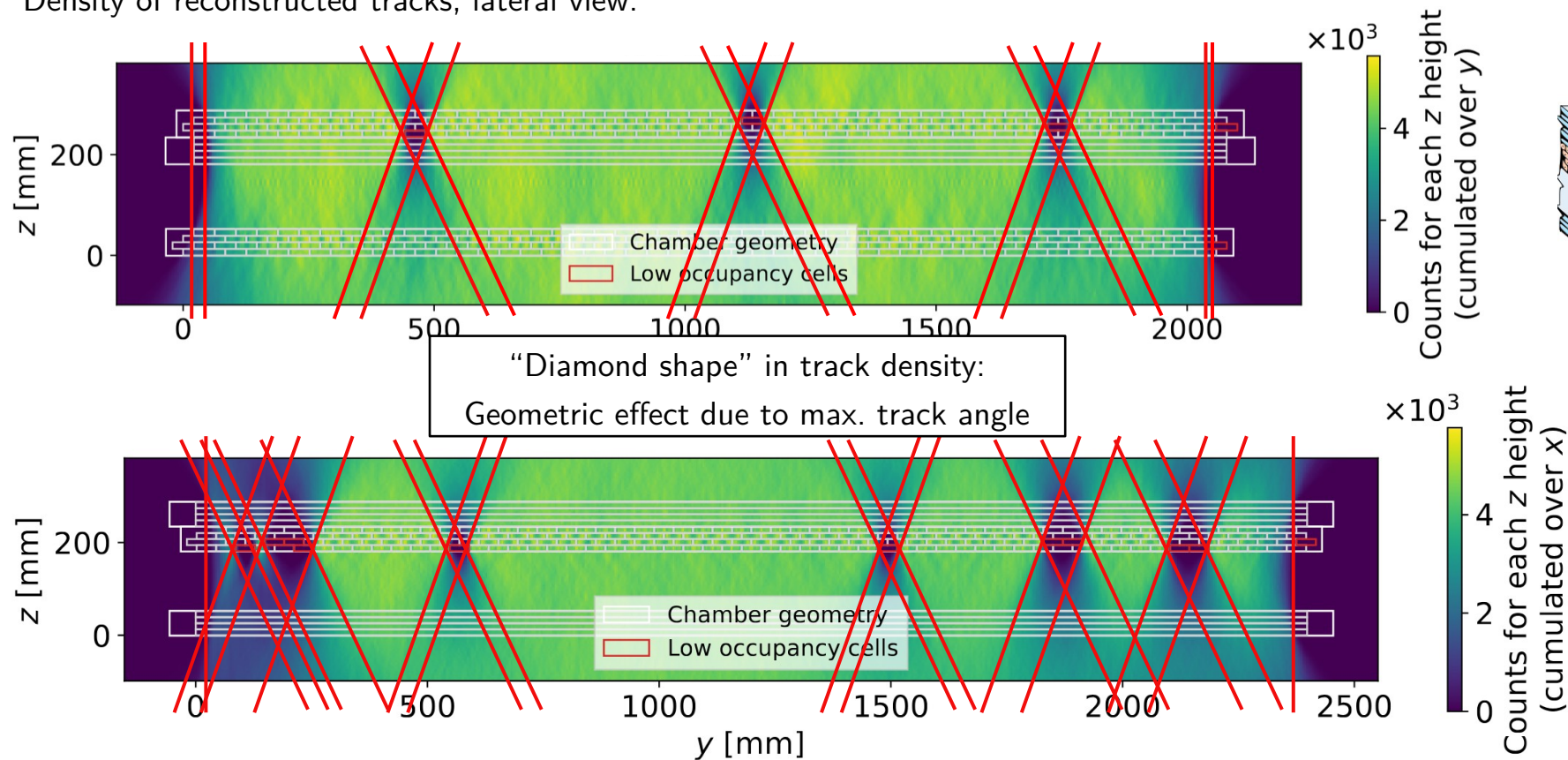
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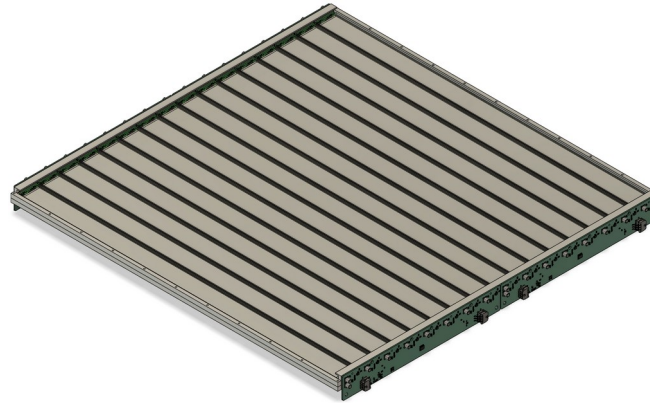
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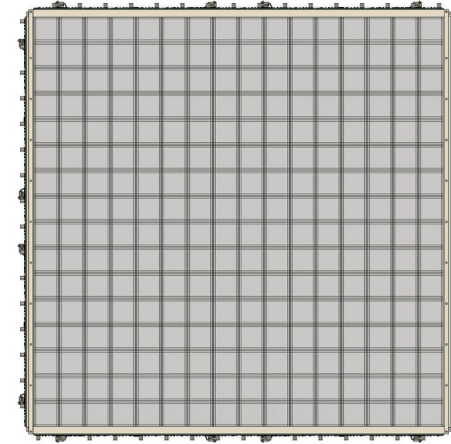
# Scintillator strip detector

# Scintillator strip detector with SiPM readout

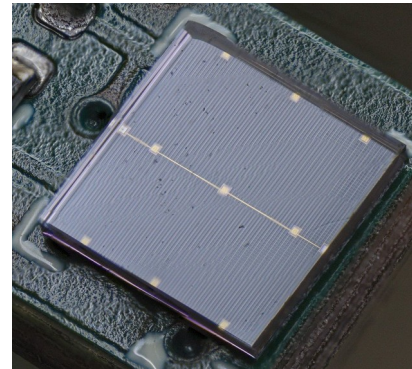
- Scintillator strip detector:
  - Developed in previous thesis projects (E. Ehlert)
  - Plastic scintillator material (EJ-200)
  - Detector size:  $50 \times 50 \text{ cm}^2$
  - Strip width: 3 cm
  - 2 layers with 16 strips each
  - Photon readout: 2 SiPMs / strip
    - 64 SiPMs in total
- Silicon photomultiplier (SiPM):
  - Solid-state detector with single-photon sensitivity
  - Array of Geiger Avalanche Photodiodes (GAPDs)
    - Single cell hit: 1 p.e. (photoelectron equivalent)
    - Gain depends on applied bias voltage



Scintillator strips



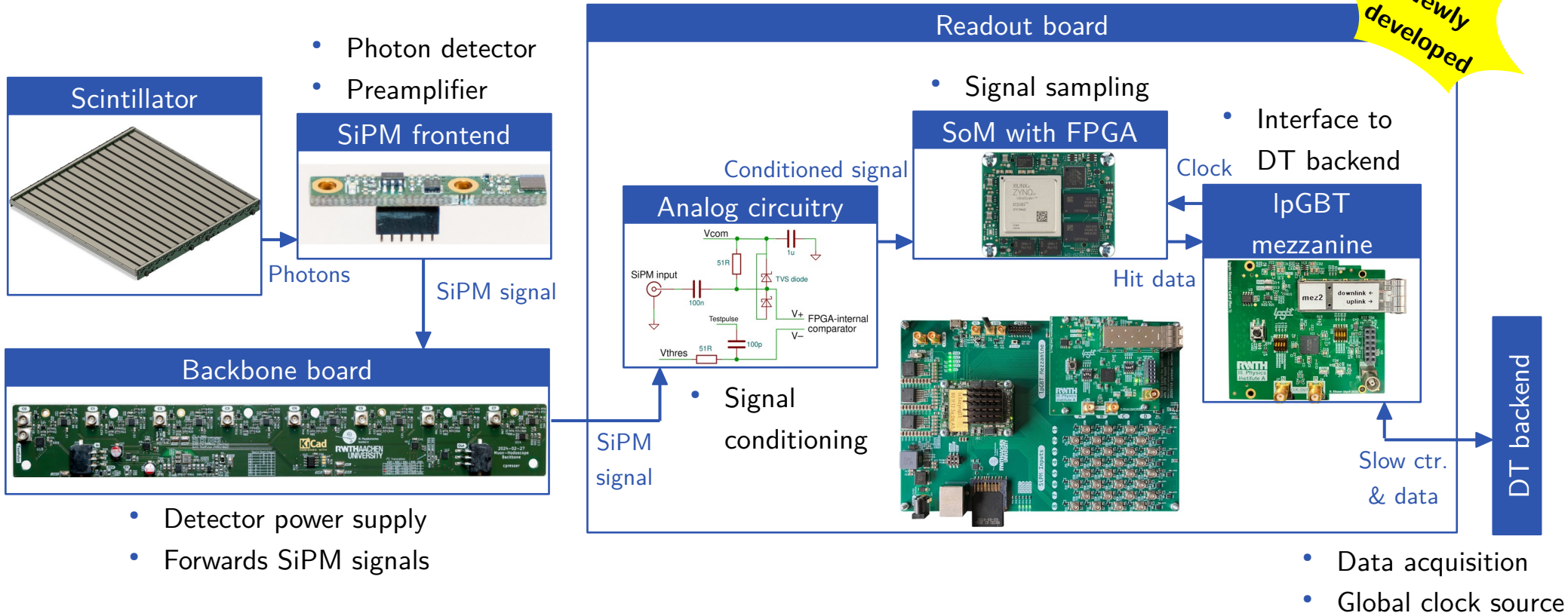
Scintillator top view



Mounted SiPM

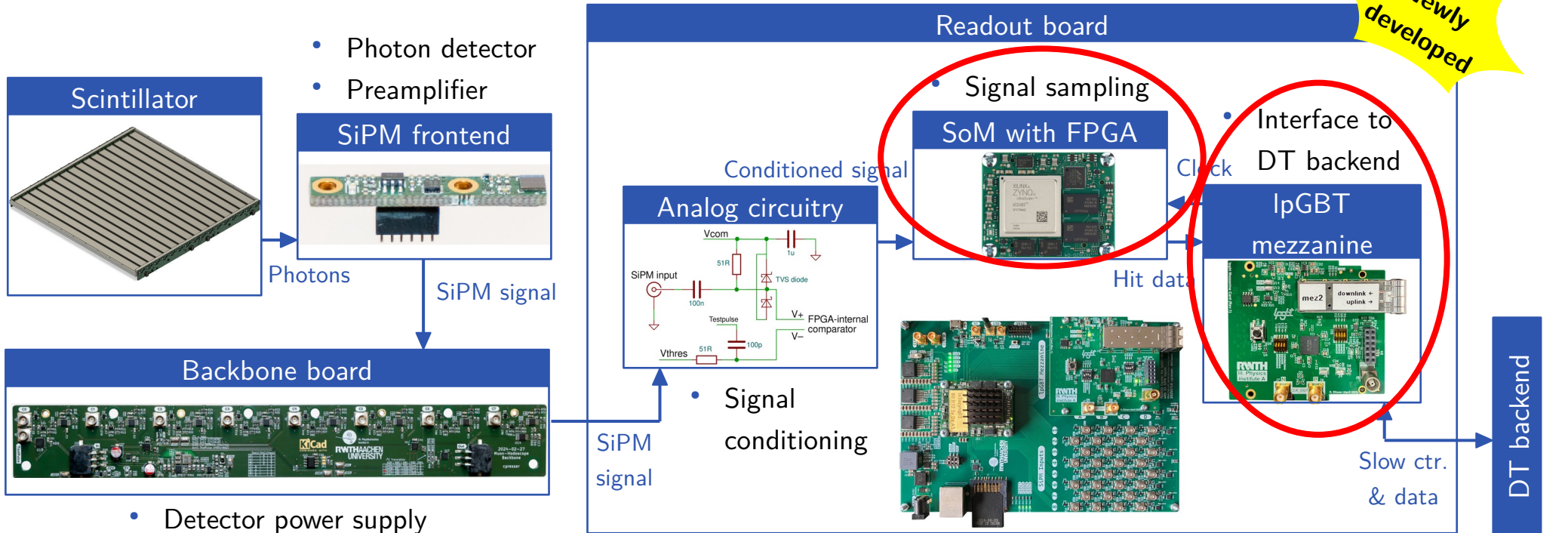
# Scintillator readout electronics – Overview

- ▶ Should have capability to interface DT backend



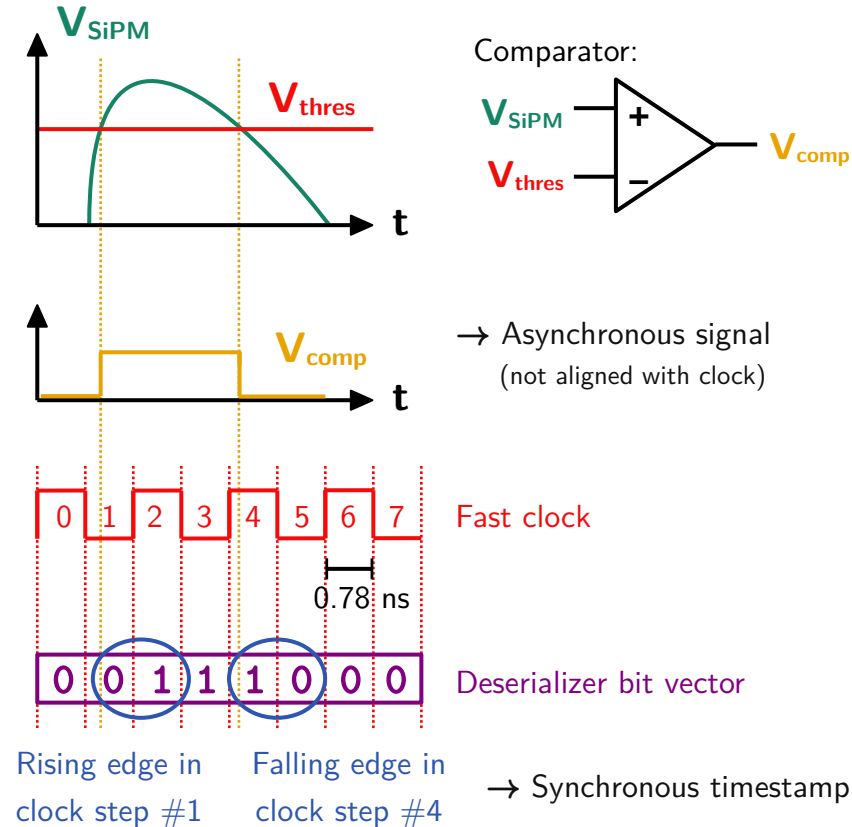
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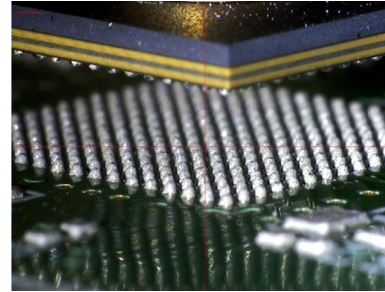
# Scintillator readout electronics – Signal sampling

- Receive preamplified analog signal from SiPM
- Discriminator:
  - Hit discrimination threshold:  $V_{\text{thres}} = 25 \text{ mV}$  ( $\sim 5$  p.e. of SiPM)
  - Implemented with FPGA-internal comparators
- Time-to-Digital Converter (TDC):
  - Assignment of digital timestamp
  - Implementation in FPGA firmware (using deserializers)
    - Deserializer returns bit vector of comparator values, sampled at each clock edge
    - Apply edge detection logic to bit vector

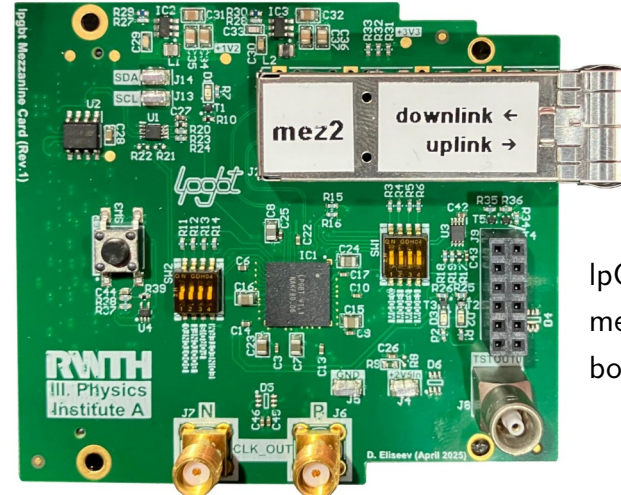


# Scintillator readout electronics – lpGBT mezzanine

- Low-Power Gigabit Transceiver (lpGBT):
  - Chip for high-speed bidirectional data transfer
  - Extensively used in Phase-2 Upgrade of CMS
  - Capability to recover synchronous clock from backend
- lpGBT mezzanine:
  - Self-developed board which hosts lpGBT chip
  - FPGA interface: Serial data lines (eLinks)
  - DT backend interface: Optical fiber



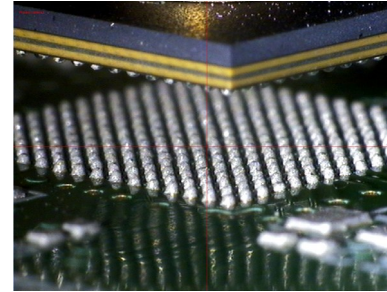
lpGBT chip being soldered onto a board



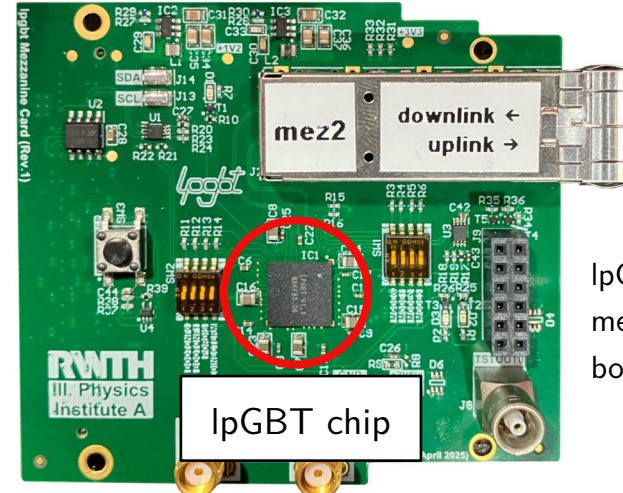
lpGBT mezzanine board

# Scintillator readout electronics – IpGBT mezzanine

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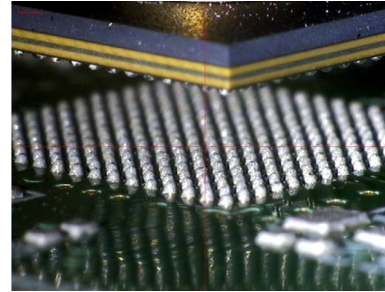
IpGBT chip being soldered onto a board



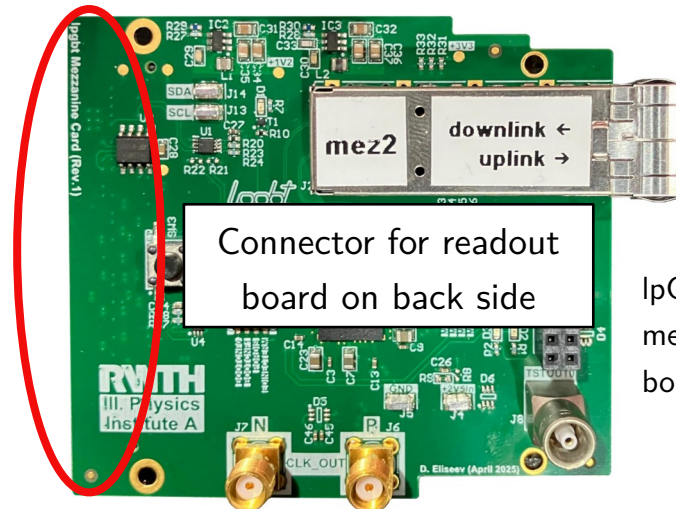
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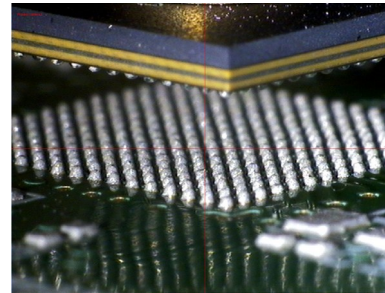
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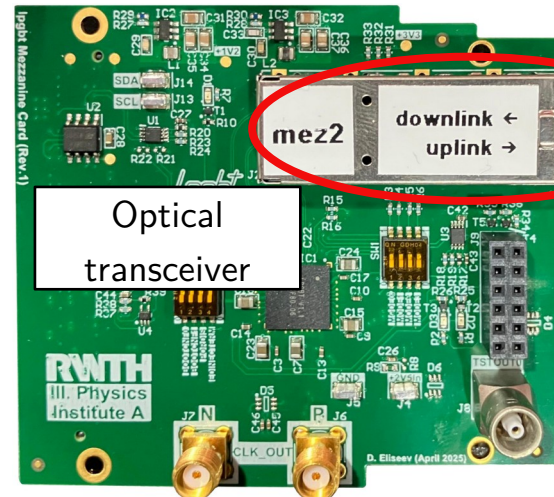
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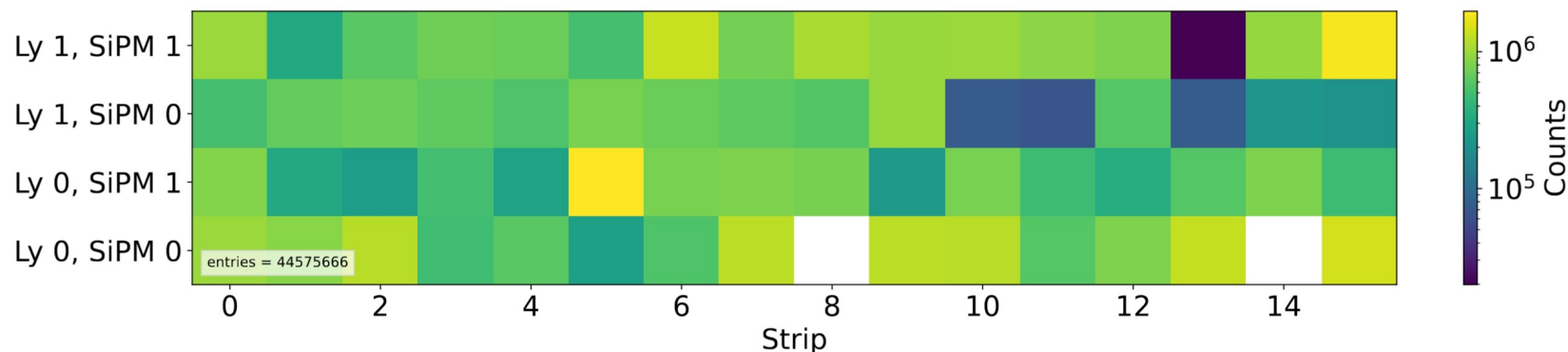
# Recorded scintillator strip detector data

# SiPM hit data

- Recorded large dataset of atmospheric muons with scintillator:
  - Suppress systematic readout effects: Apply dead time of 250 ns

| Duration | SiPM hits  |
|----------|------------|
| 5h 27min | 44,575,666 |

SiPM occupancies:



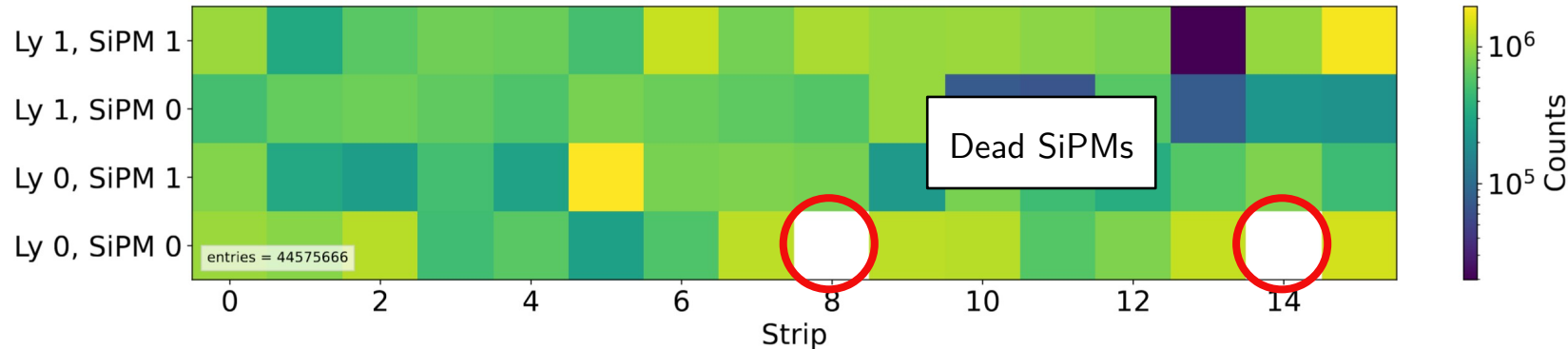
- Average SiPM rate: **37 Hz**
  - Rate fluctuations over one order of magnitude
    - Common bias voltage for all SiPMs, i.e. different effective p.e. thresholds due to fabrication tolerances
- 2 dead SiPMs** in detector

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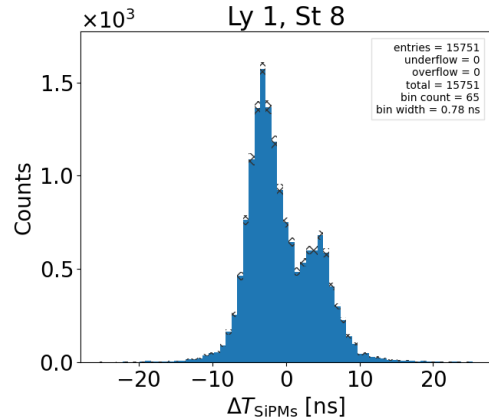


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# Scintillator strip coincidences

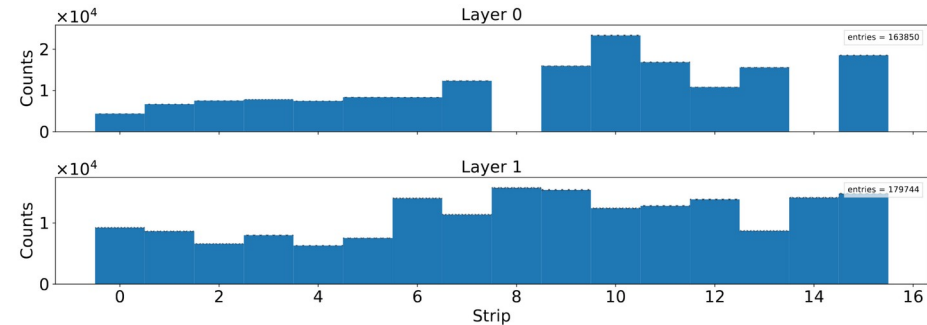
- Evaluate SiPM coincidences for scintillator strips
  - Coincidence window:  $\Delta T_{\text{SiPMs}} \leq 25 \text{ ns}$
  - Require isolation from other hits in layer

Time difference  $\Delta T_{\text{SiPMs}}$  of coincidence:



- ▶ **Contained coincidence timing  $O(15 \text{ ns})$** 
  - ▶ Indicates “real” coincidences
- ▶ Double-peak structure observed
  - ▶ Needs further investigation, possibly due to sampling logic

Obtained strip occupancies:



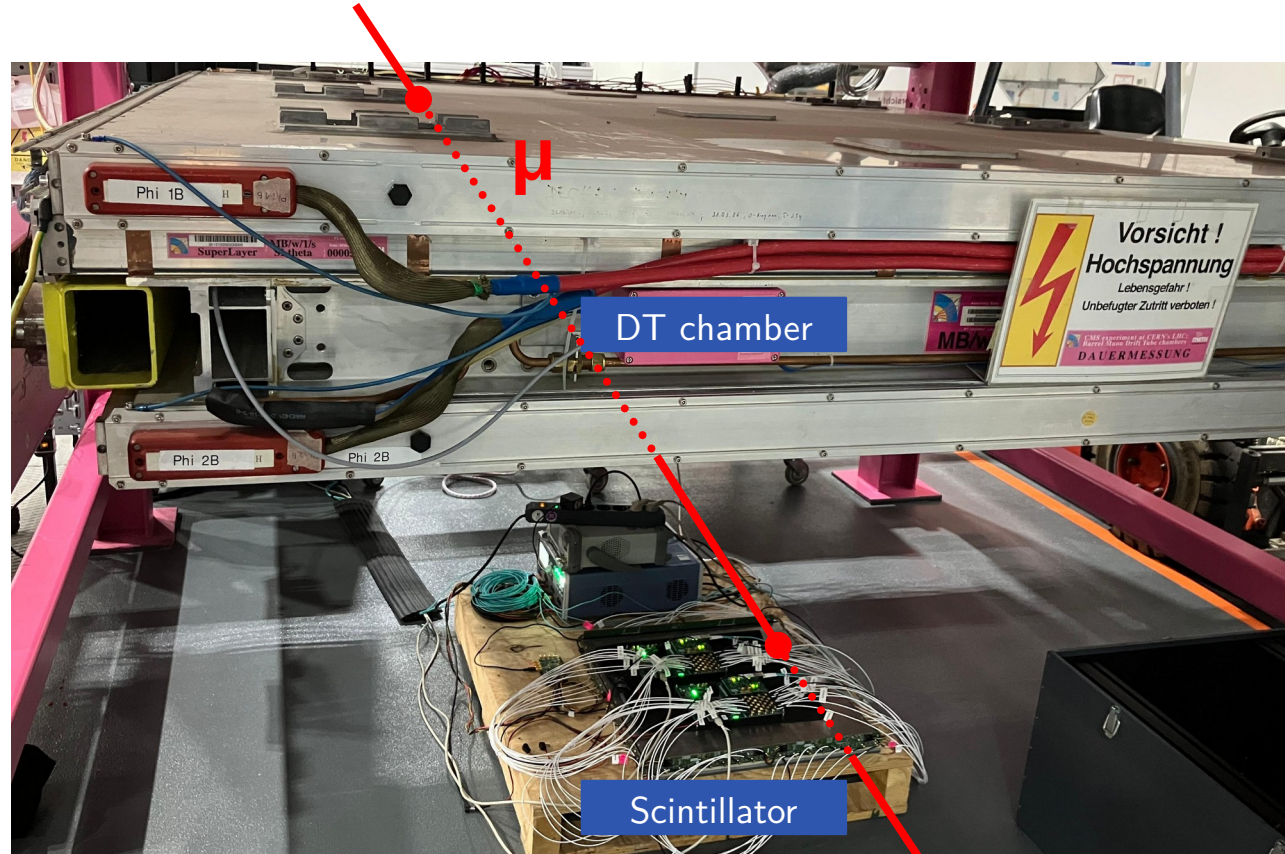
- ▶ Measured strip rate:  **$O(0.5 \text{ Hz})$** 
  - ▶ Observe strip rate fluctuations
    - SiPM bias voltage calibration should be performed
- ▶ **2 dead strips** from dead SiPM propagation

Simplified atmospheric muon simulation: **2.2 Hz**

# Combination of detector data

# Combined measurement setup

- Place scintillator below DT chamber at cosmic test stand
- ▶ **Attempt combined data analysis of reconstructed DT tracks & scintillator strip hits**



# Temporal matching of DT tracks & scintillator hits

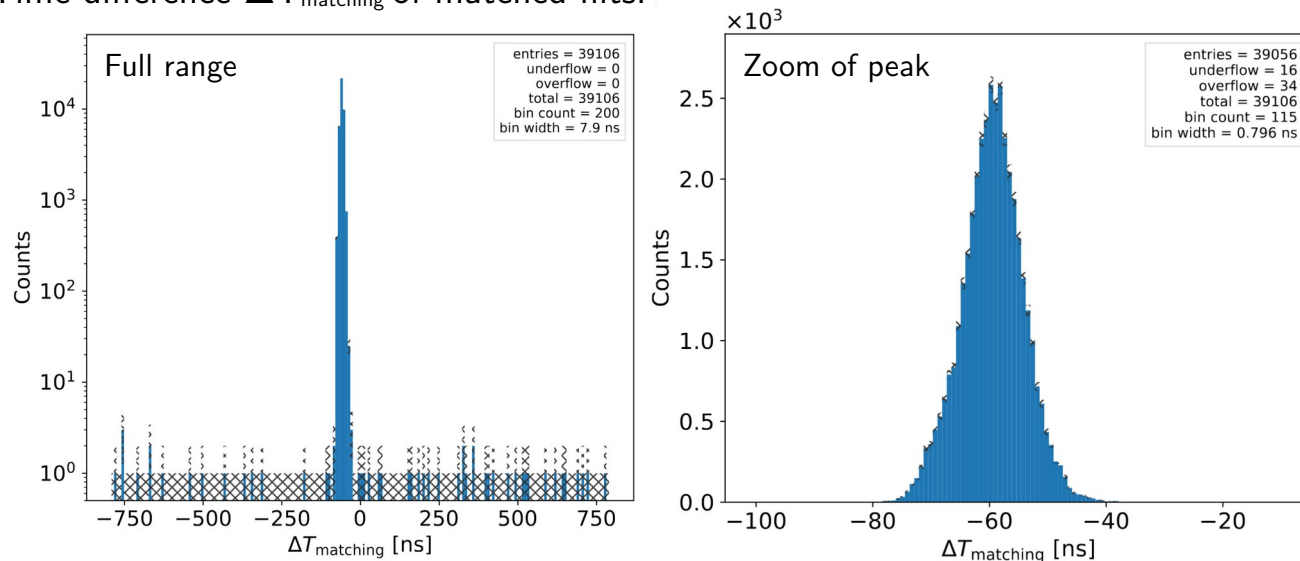
- Matching of detector hits by arrival time:
  - Require DT track & scintillator strip hit coincidence within 780 ns:
$$\Delta T_{\text{matching}} = T_{\text{scint}} - T_{\text{DT}}$$
  - Remove ambiguities while matching

- ▶ Matched hit rate: **2 Hz**

For comparison:

- Scaled DT track rate: 4 Hz
- Scintillator layer rate: 8 Hz

Time difference  $\Delta T_{\text{matching}}$  of matched hits:



- ▶ **Successful detector synchronization !**

- ▶ **Peak FWHM: 12 ns**

- ▶ Includes all timing uncertainties of readout & reconstruction

- ▶ Observe constant **latency offset**

- ▶ DT tracks are -59 ns later than scintillator hits

- ▶ Possibly due to readout propagation difference

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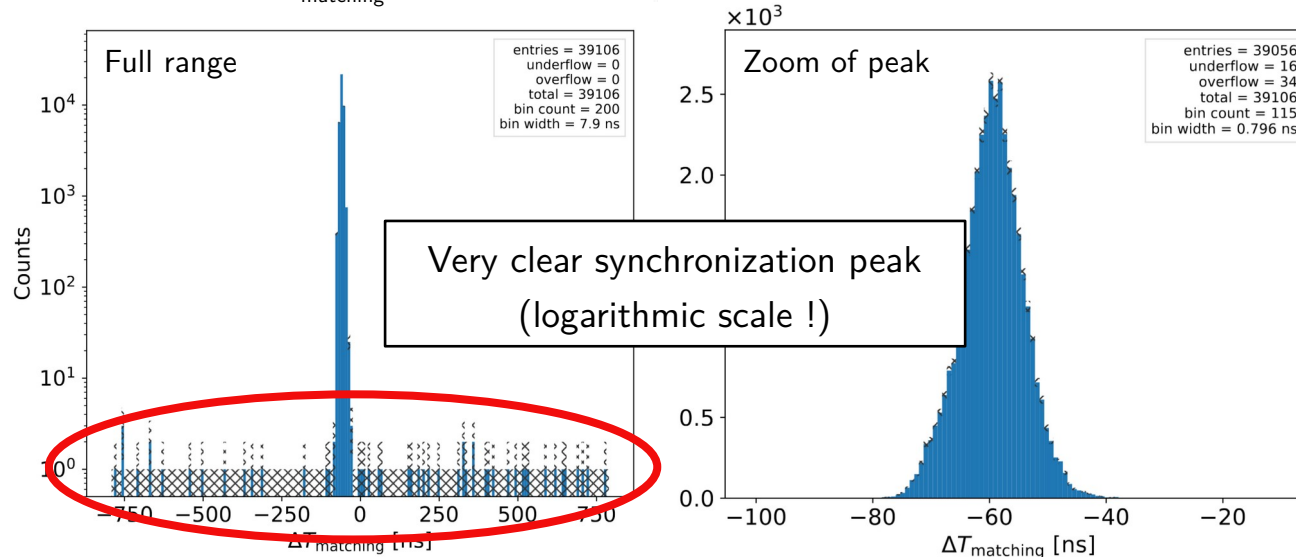
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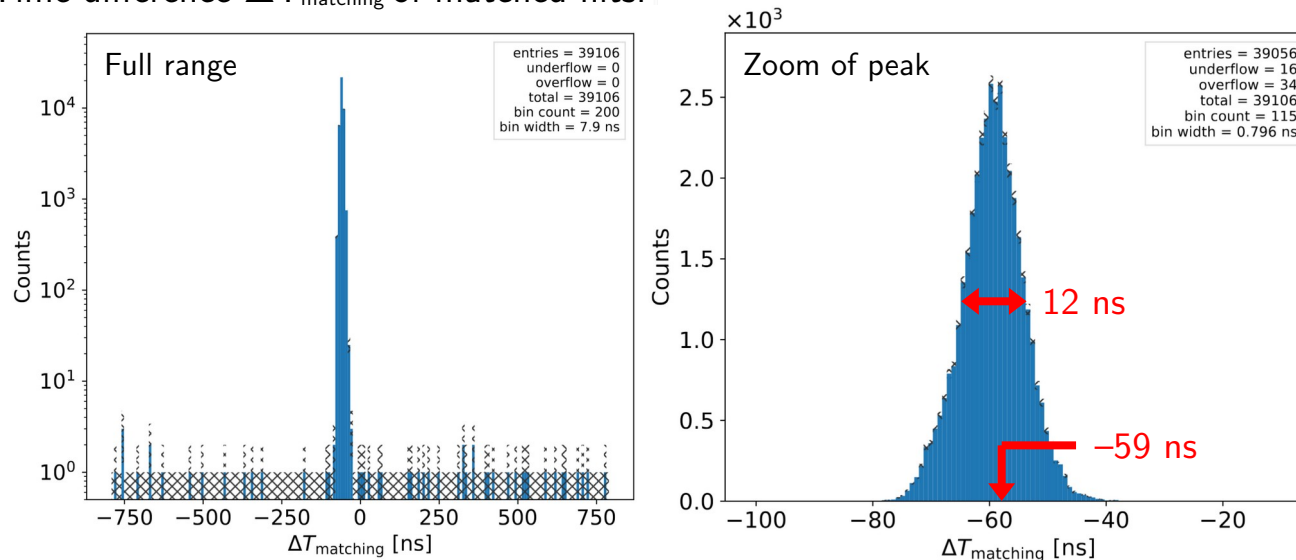
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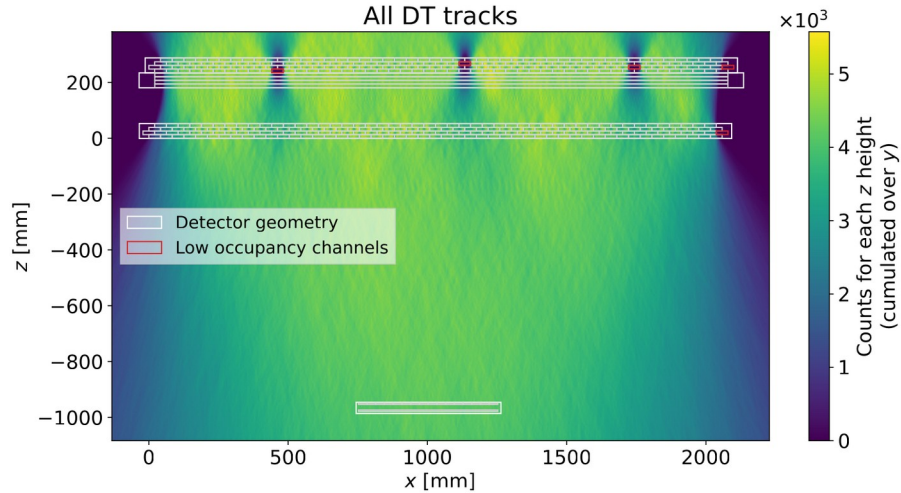
Time difference  $\Delta T_{\text{matching}}$  of matched hits:



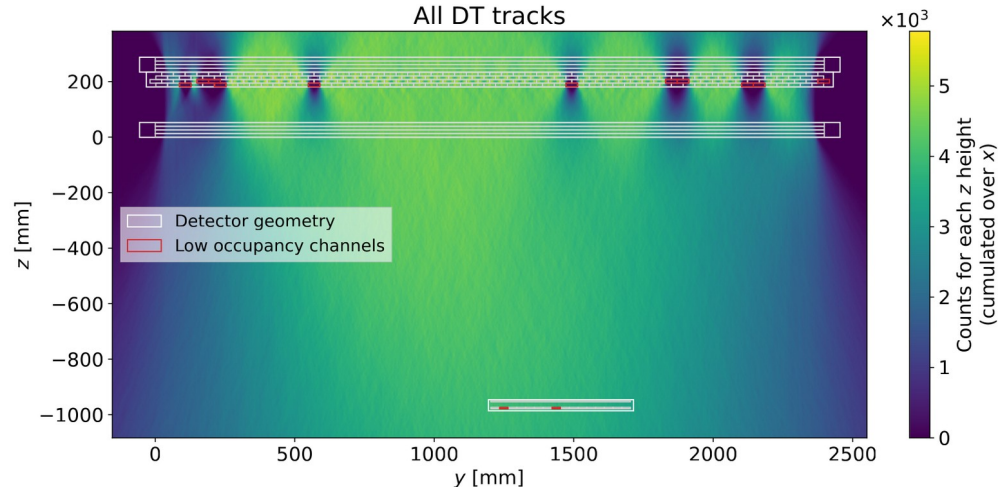
► **Successful detector synchronization !**

- **Peak FWHM: 12 ns**
- Includes all timing uncertainties of readout & reconstruction
- Observe constant **latency offset**
  - DT tracks are -59 ns later than scintillator hits
  - Possibly due to readout propagation difference

# Independent cross-check of DT track reconstruction

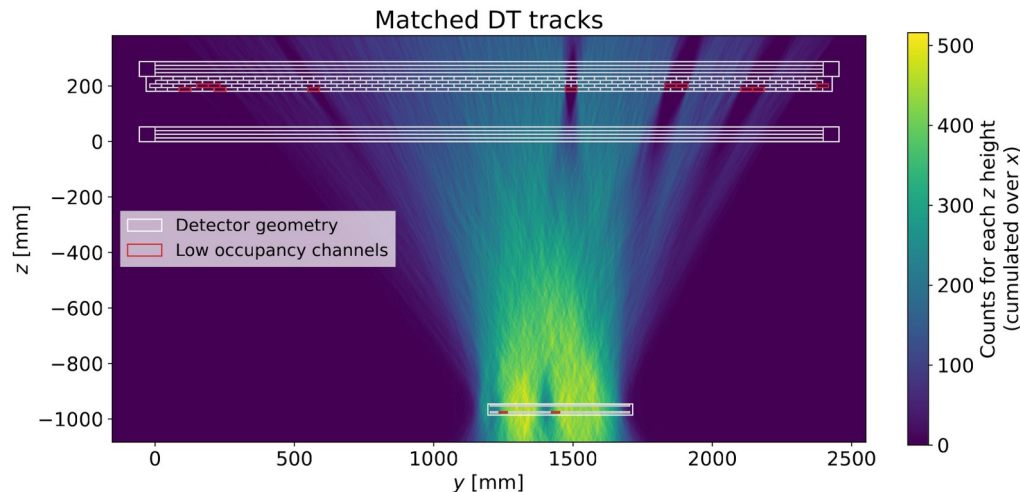
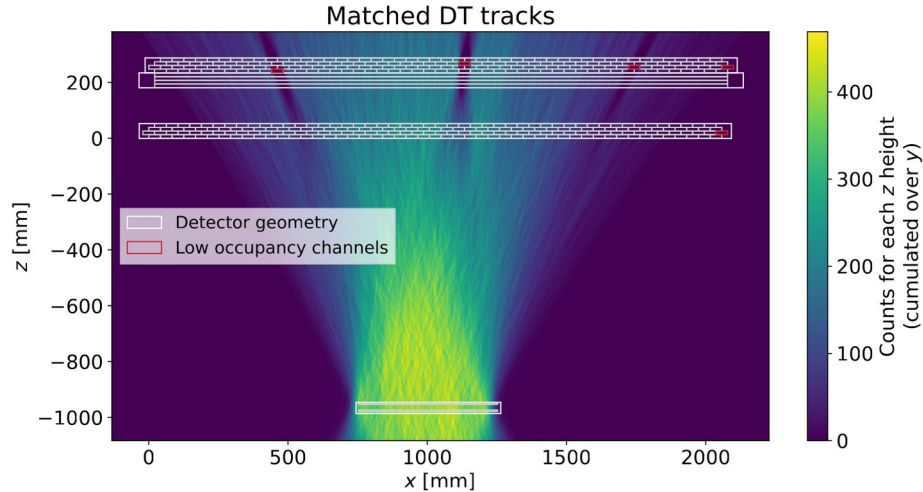


- Cross-check of DT track reconstruction
  - Analyze position of matched tracks w.r.t. scintillator
  - Unbiased, as tracks were only matched in time



Density of matched  
reconstructed tracks,  
in lateral view.

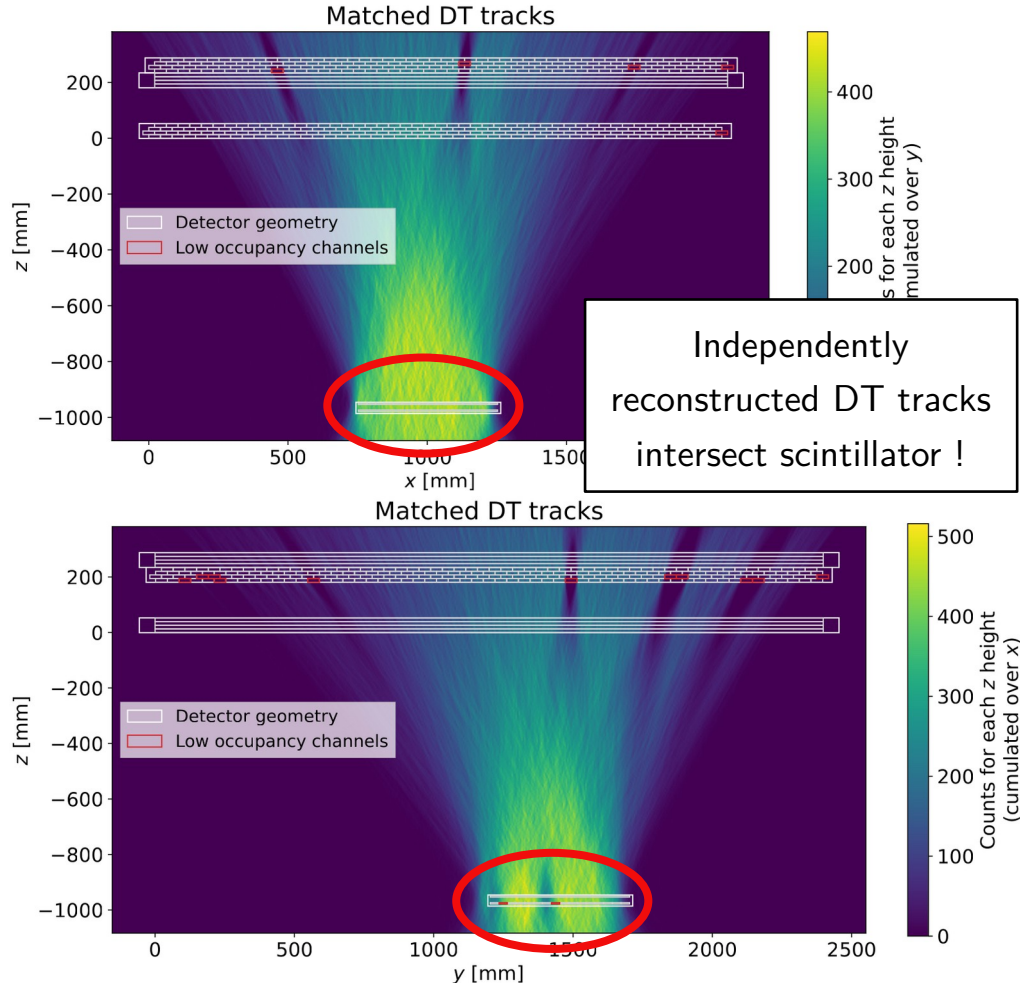
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- ▶ **Matched tracks are intersecting scintillator as expected !**
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  - ▶ Visible effect of DT cells & scintillator strips
- ▶ Need for proper **detector alignment**

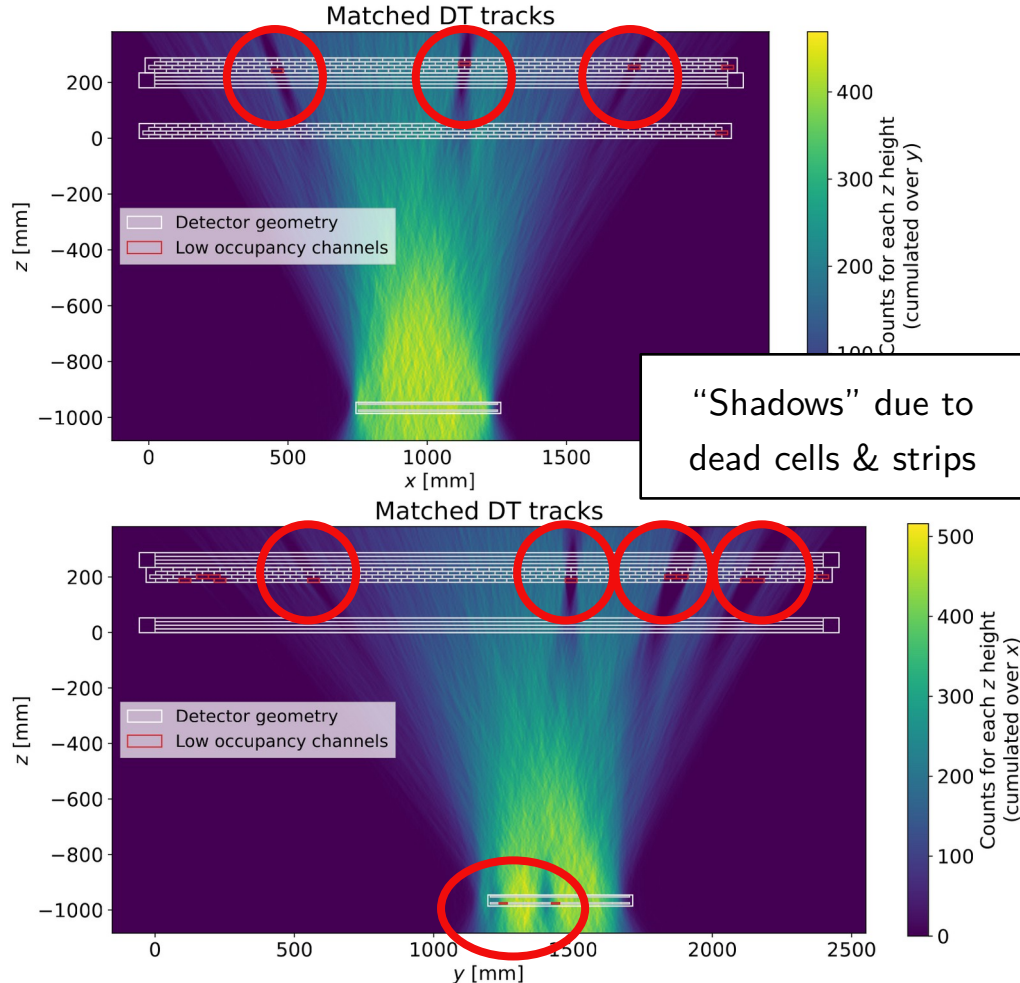
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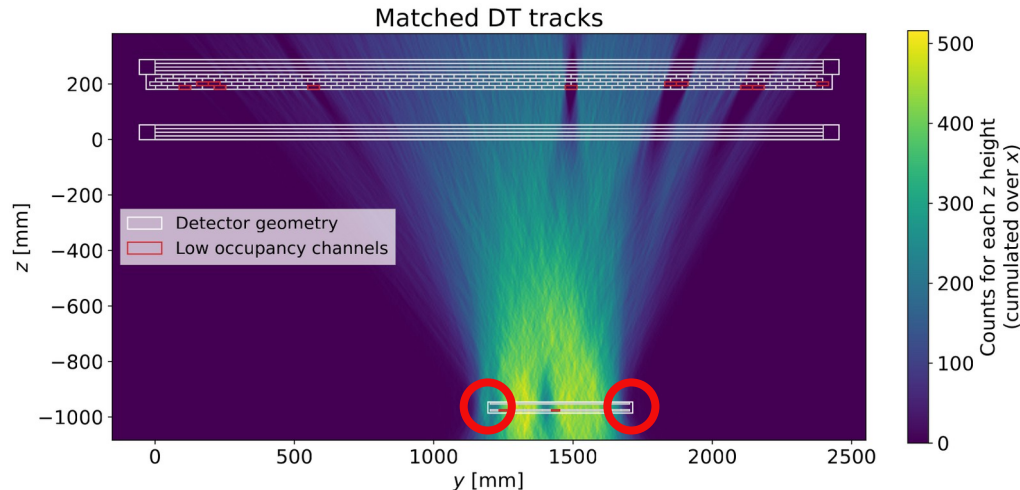
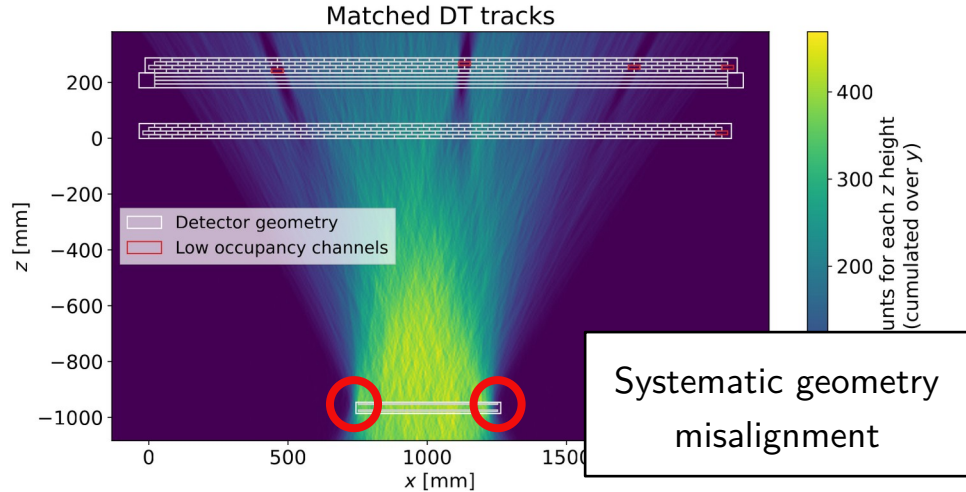
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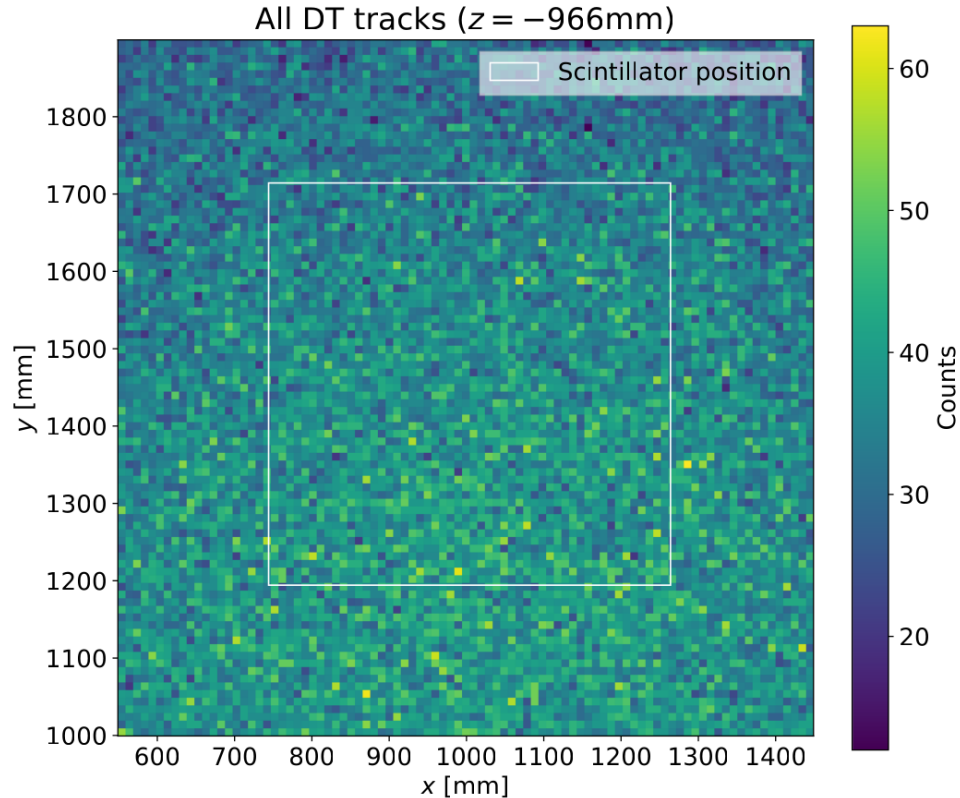
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- Density of matched reconstructed tracks, in lateral view.

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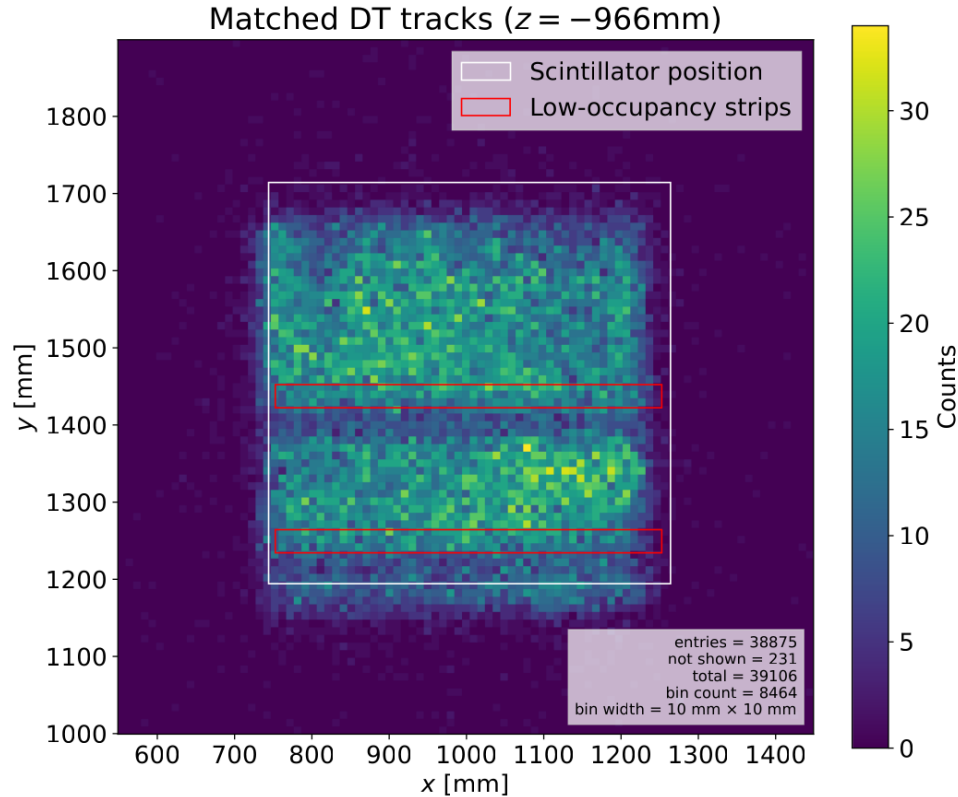
2D histogram of matched tracks in scintillator:



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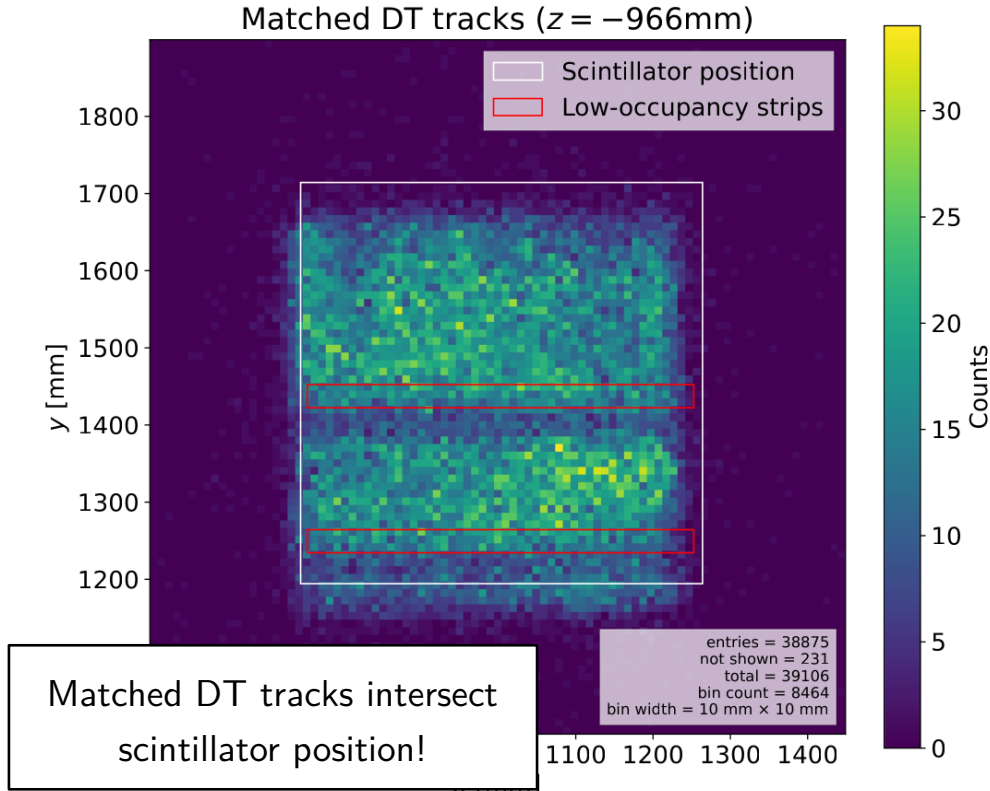
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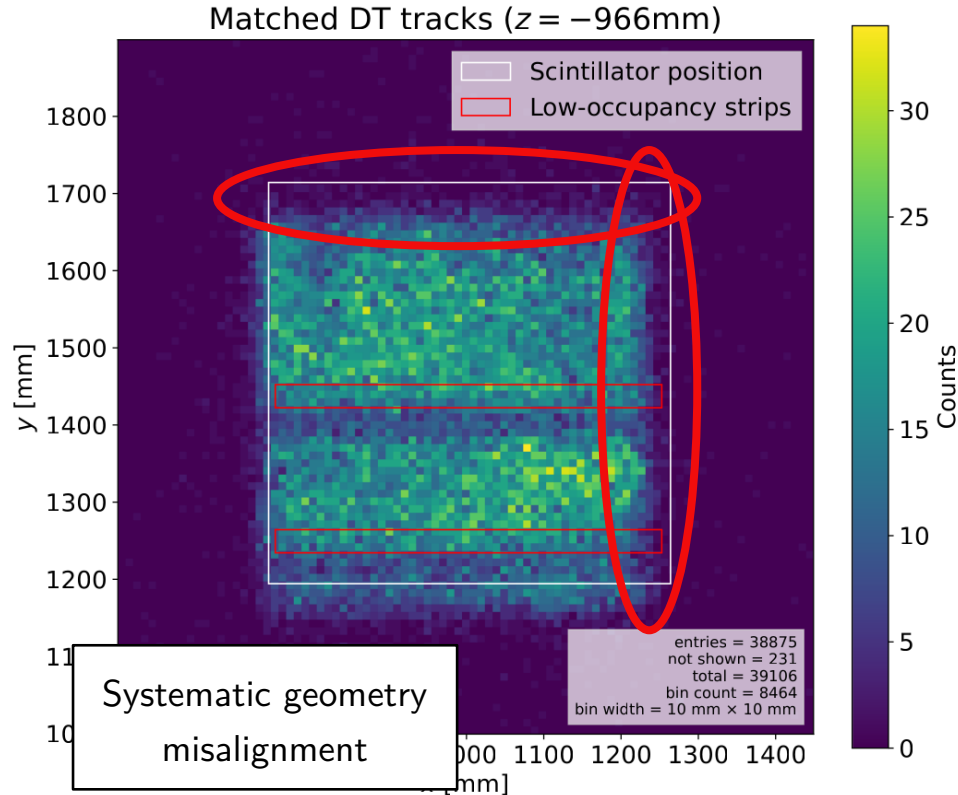
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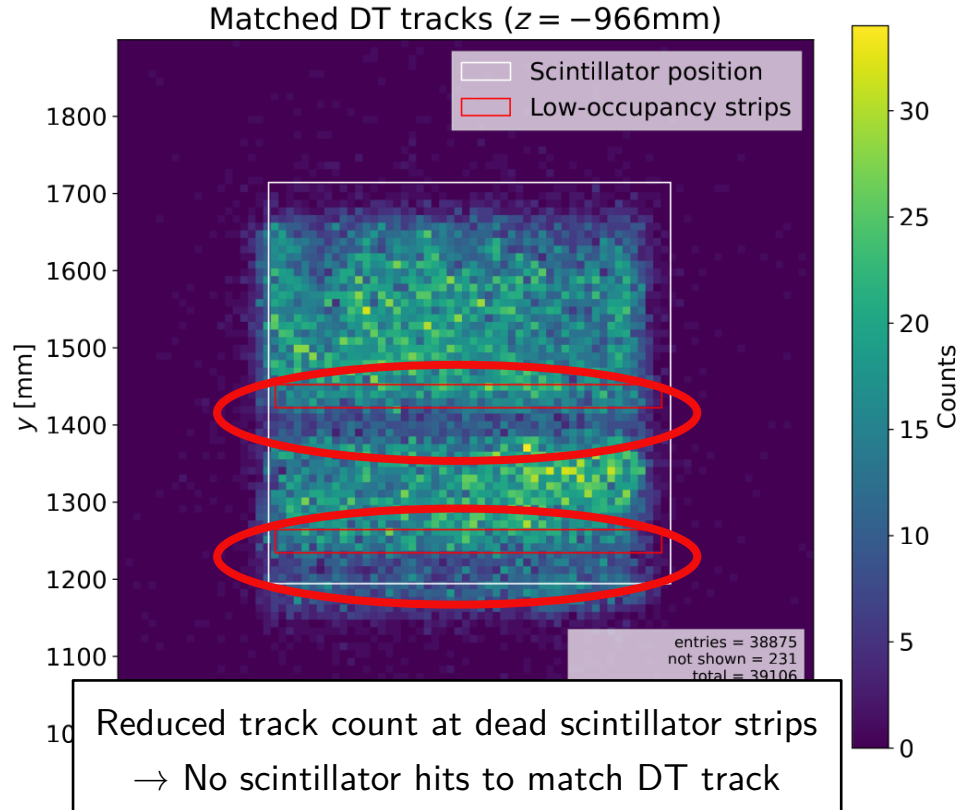
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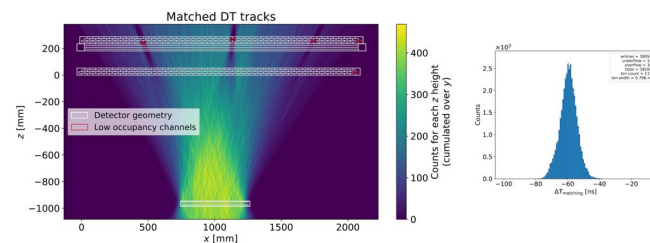
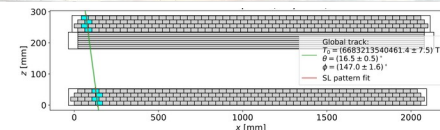


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# Summary & outlook

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- DT chamber at cosmic test stand:
  - Successful installation & commissioning of CMS DT Phase-2 electronics
  - Implementation of triggerless track reconstruction algorithm
- Scintillator strip detector:
  - Developed new readout system & firmware (featuring IpGBT chip)
  - Observed systematic effects in SiPM readout and established mitigation with dead time
- Combined analysis:
  - Detector synchronization  $O(10\text{ ns})$  was achieved, including all timing uncertainties
  - Positive independent cross-check of DT track reconstruction



## Outlook:

- ▶ Further investigate systematics in scintillator readout system
- ▶ Optimize triggerless DT reconstruction algorithm & investigate possibility to use online DT trigger algorithm
- ▶ Provide detector alignment procedure
- ▶ Gained experience can be used to convert DT chamber to cosmic test stand for other detectors (beyond the scintillator)

# Thank you for your attention!

Questions and comments are of course welcome.

# Thank you!

I would like the following (incomplete) list of people who made this thesis possible:

- **Prof. Dr. Alexander Schmidt** for the opportunity and support to write my thesis at his institute
- **Apl. Prof. Dr. Oliver Pooth** for kindly agreeing on being the second corrector
- **Dr. Markus Merschmeyer** for his support and valuable feedback
- **Dr. Dmitry Eliseev** for supervising this thesis and his tireless support
- **Carsten Presser** for contributing several important aspects
- **Prof. Dr. Thomas Hebbeker** for his interest in my work and the helpful suggestions
- (...)

# Backup

- 1) Theory
- 2) DT chamber description & data analysis
- 3) Scintillator description & data analysis
- 4) Combined data analysis
- 5) Simplified Monte Carlo simulation of cosmics

## Backup (Pt. 1)

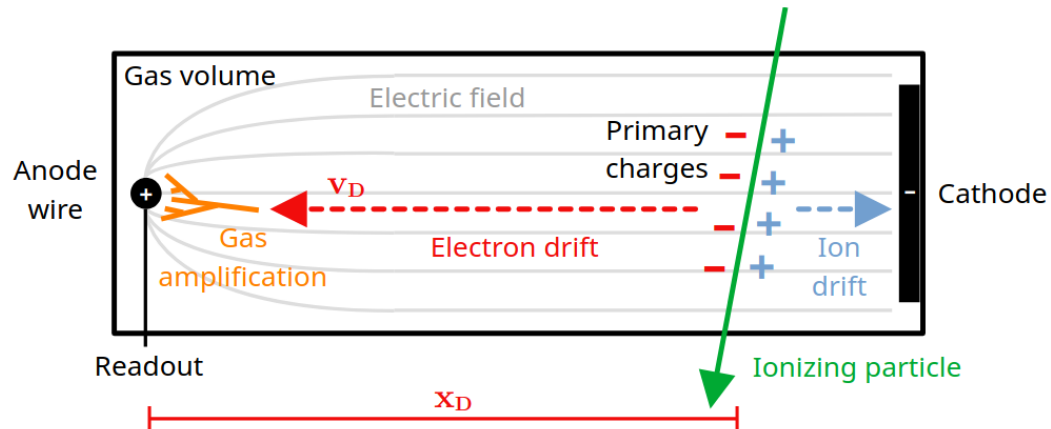
- Theoretical background
- Principles of particle detection

# Atmospheric muons

- Cosmic rays interact with Earth's atmosphere
- Muons originate from decay of light hadrons
- Vertical muon flux:  $\Phi = 70 \text{ m}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$
- Zenith angle dependence:  $\Phi(\theta) \sim \cos^2 \theta$

# Drift chamber

- Ionization of gas in detector volume from passing muon
- Separation of charges in electric field
- Charge drift with constant velocity
- Gas amplification in proximity of wire
- Initial hit position can be reconstructed from drift time measurement
  - Resolution far beyond the drift cell granularity

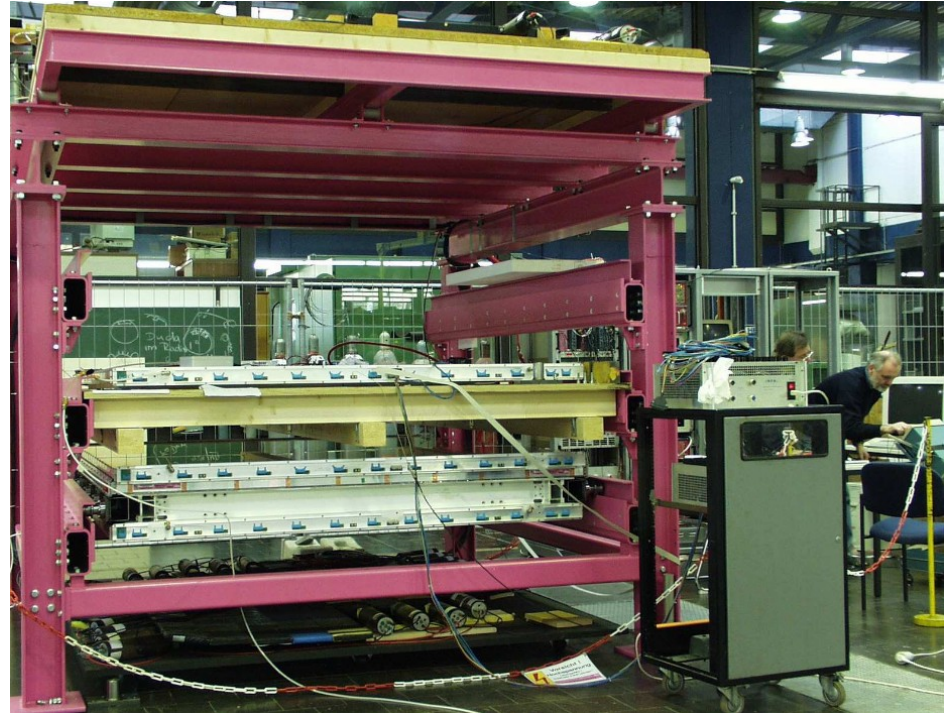


## Backup (Pt. 2)

- CMS DT system
- Triggerless DT data analysis

# Cosmic test stand in early 2000s

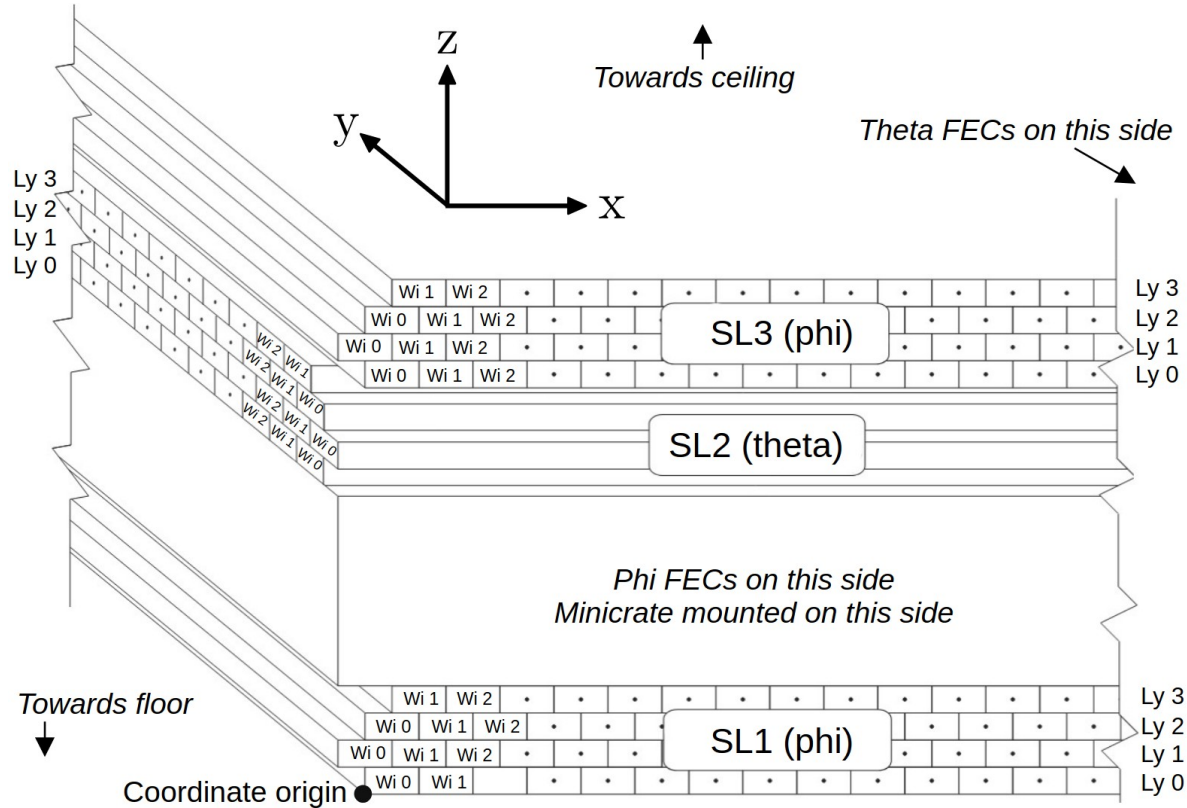
Used for quality control of DT chamber production at our institute:



- ▶ Trigger scintillators above & below chamber
- ▶ Custom TDCs for drift time measurement

# DT global coordinate frame

Selected global coordinate frame in this work:



# DT readout electronics

DT chamber frontend board:

- Signal discrimination: 35 mV threshold
- Output of digital hit pulse

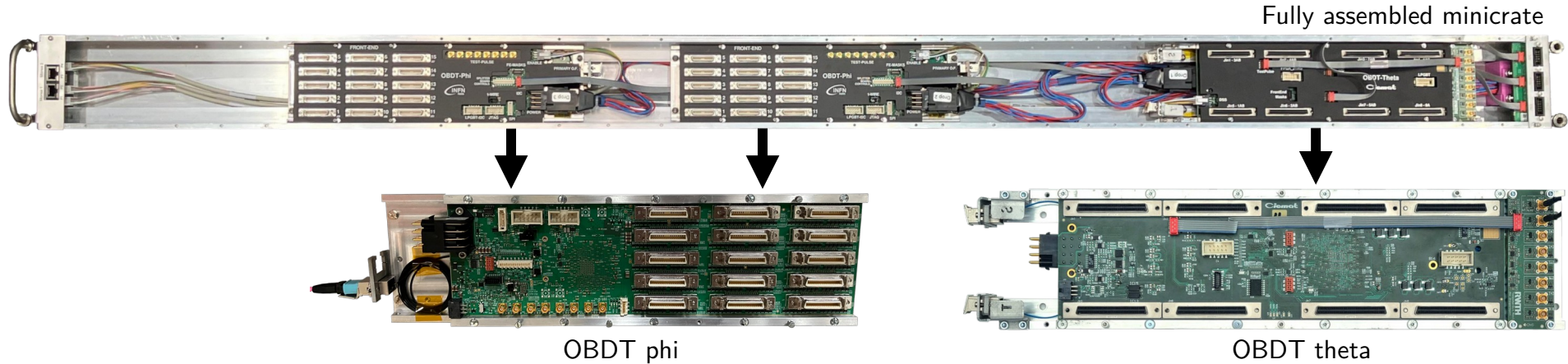


Frontend board

# DT readout electronics

Minicrate (on-chamber readout electronics):

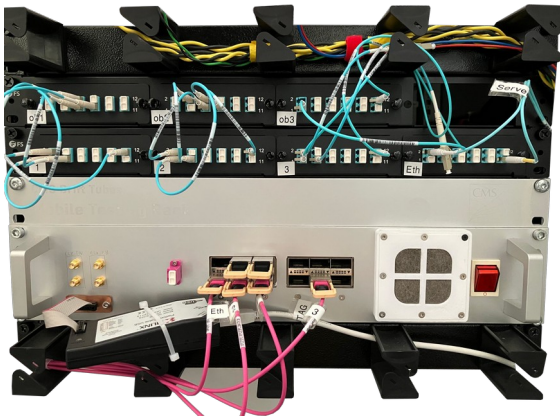
- On-board Electronics for Drift Tubes (OBDT) boards in two flavors (phi & theta)
  - Hit digitization with 0.78 ns granularity
  - Data transmission over 10.24 Gb/s optical links



# DT readout electronics

## Backend system:

- Prototype backend hardware: “Slow control & timing box” (SC&T box)
  - No online trigger algorithms, only stores raw DT hits
- Actual CMS DT Phase-2 backend system not available yet

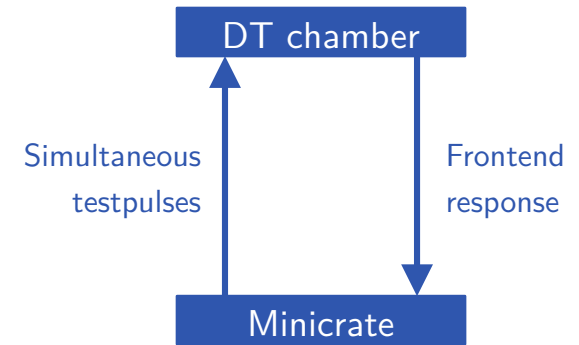
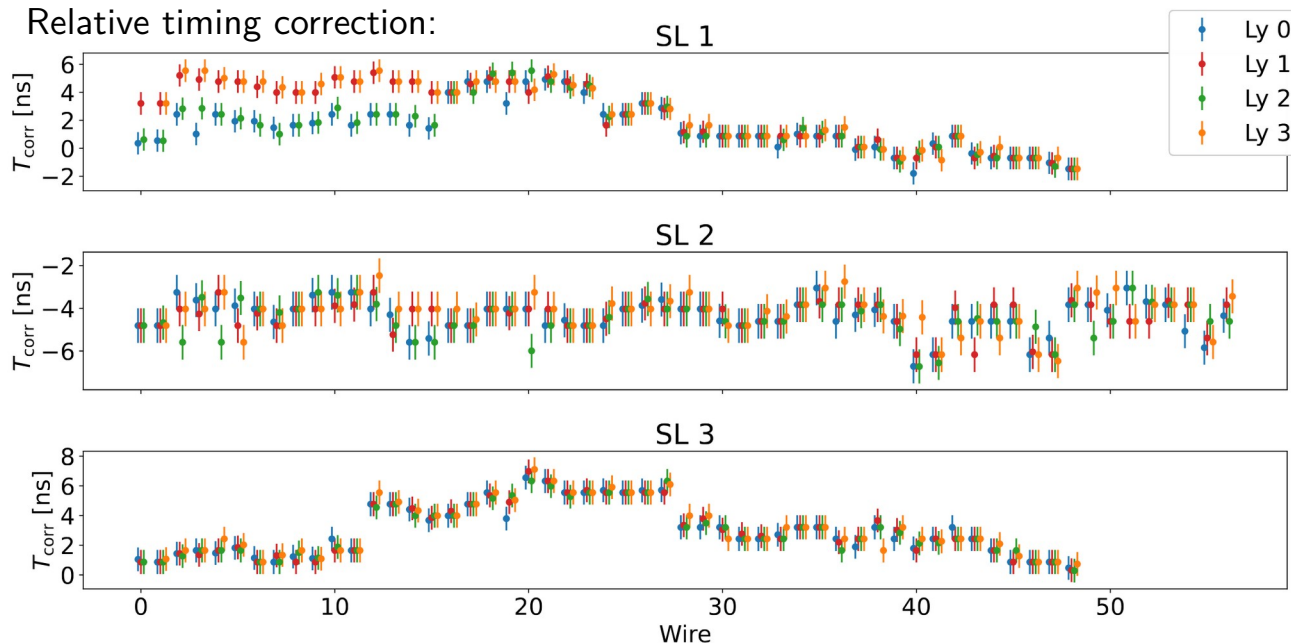


Optical fiber patch panel

SC&T box backend

# DT timing calibration with testpulses

- DT reconstruction relies on cell hit timestamps with  $O(\text{ns})$  precision
  - Relative timing calibration is necessary
- Minicrate generates simultaneous testpulses
  - Well-defined timing of sent testpulse
  - Record response timestamp and estimate relative correction  $T_{\text{corr}}$

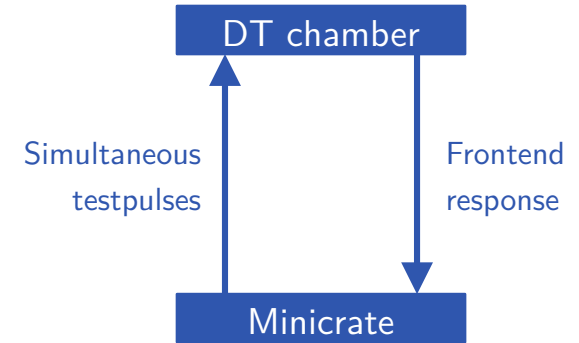
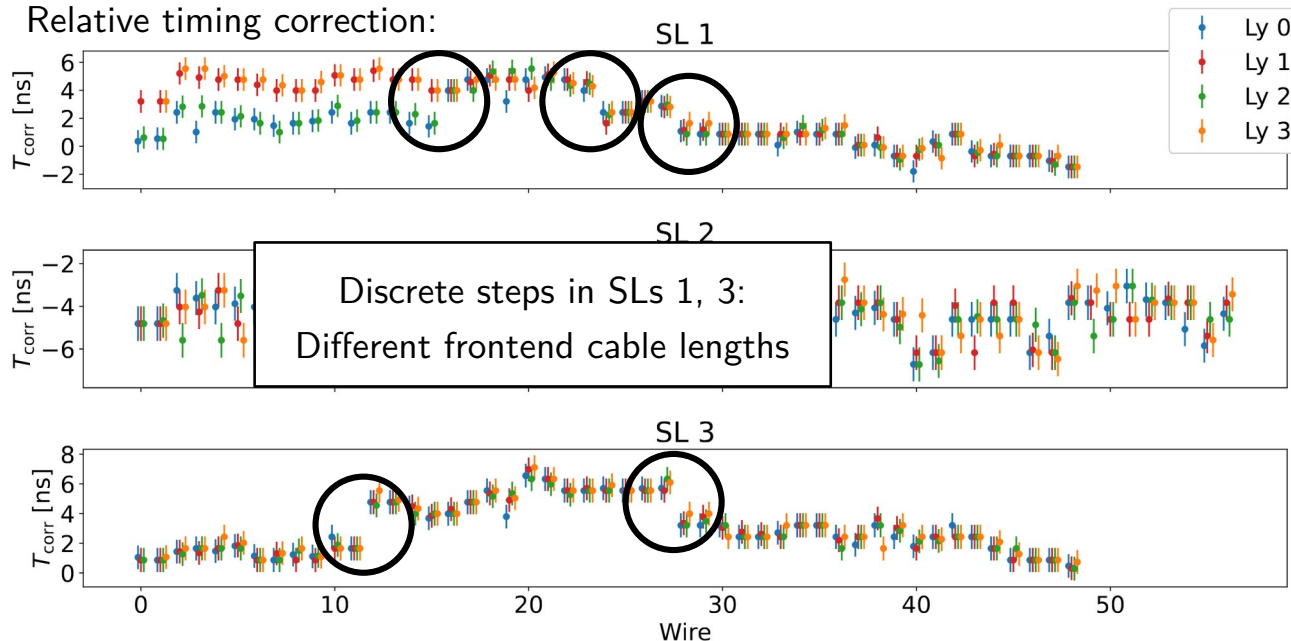


$$T_{\text{corr}} = T_{\text{response}} - \langle T_{\text{response}} \rangle$$

- ▶ Obtain DT timing corrections in the expected range ( $< 10$  ns)
- ▶ Are applied in offline data analysis

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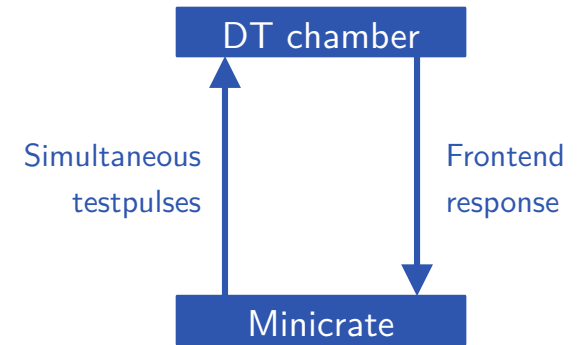
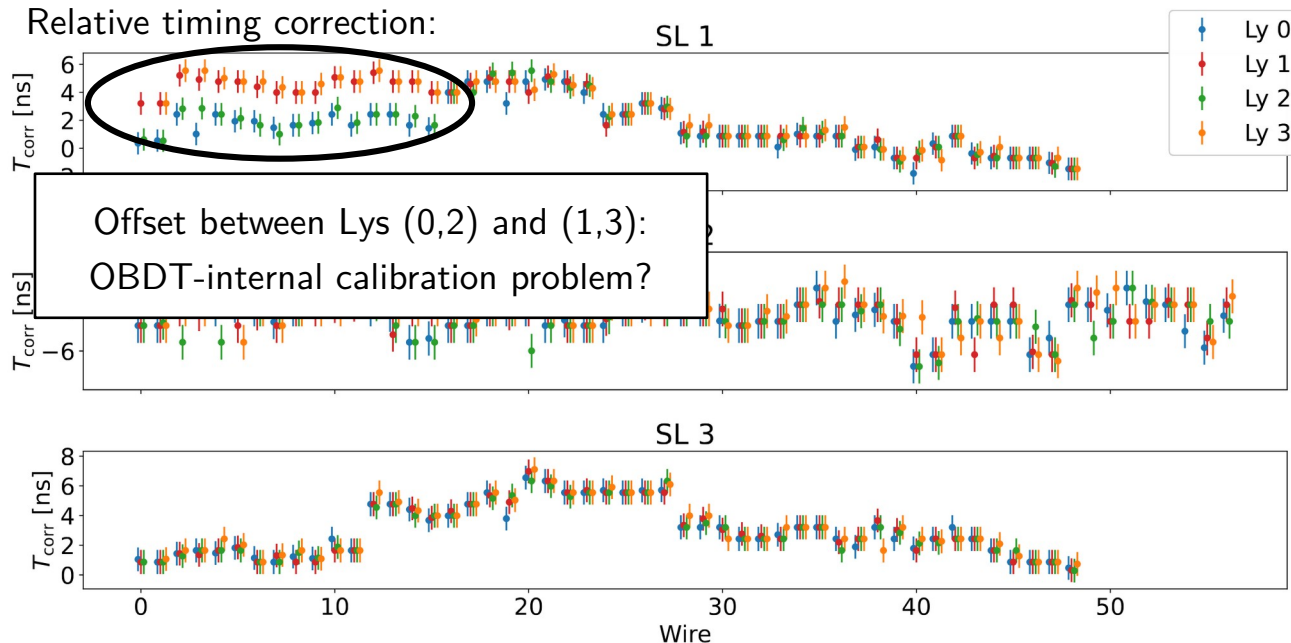


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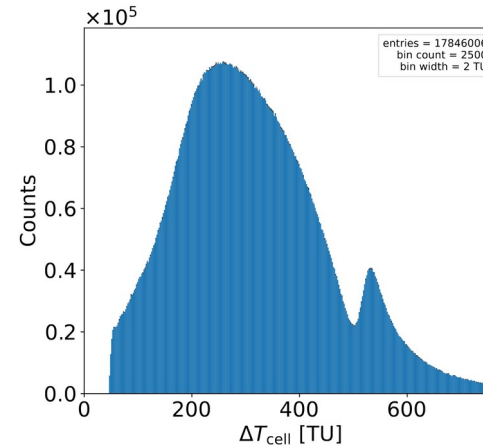
# DT secondary hits

- Time difference between consecutive hits of same cell reveals presence of secondary hits
  - Remove Poisson background events
  - Observe double peak structure in  $< 500$  ns range
- Similar structure was observed in CMS publications

Physical cause:

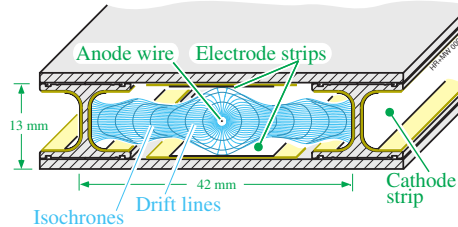
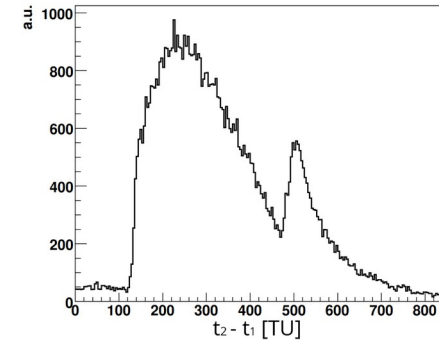
- Photoeffect leads to secondary ionization
  - Photons from avalanche near anode wire create photoelectrons at electrodes
    - Peak 1: Field shaping electrodes
    - Peak 2: Cathode electrodes ( $t_{\text{drift}} \approx t_{\text{drift,max}}$ )

Time difference between hits:



1 TU (timestamp unit) = 0.78 ns

Comparison with  
CMS-NOTE-2008-003:



- Apply individual cell dead time of 468 ns to suppress secondary hits in the recorded dataset

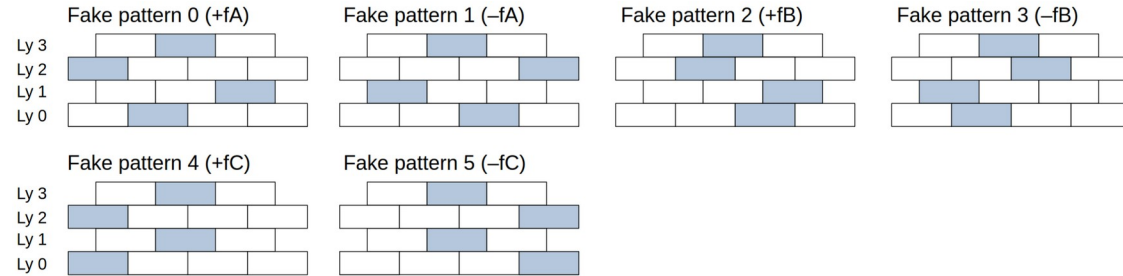
Estimation of drift velocity:  $\mathbf{v_{drift} = d_{half-cell} / (t_{peak\ 2} - t_{initial}) \approx (50.5 \pm 0.2) \mu m/ns}$

- Doubts about precision: Need detailed simulation to understand all involved effects
- Therefore continue analysis with CMS reference value:  $v_{\text{drift}} = 54.5 \mu\text{m/ns}$

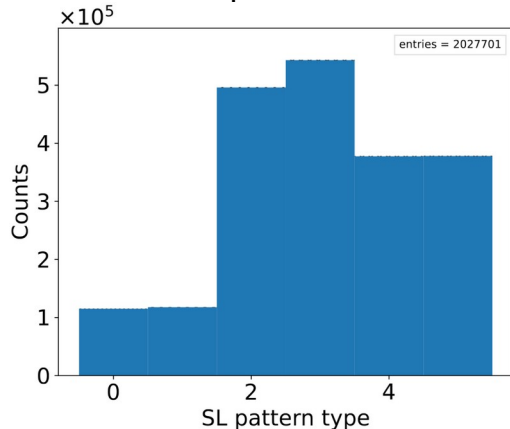
# DT superlayer fake patterns

- Analyze influence of noise hits with fake patterns

## Superlayer fake pattern geometry:



## Pattern occupancies:

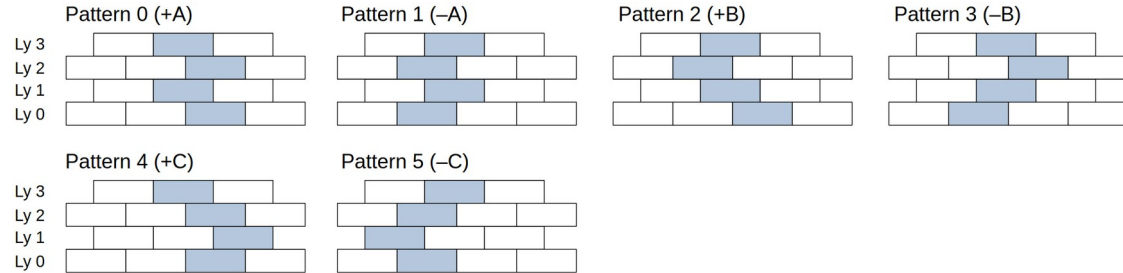


- Observe cumulated pattern rate of 35 Hz / superlayer

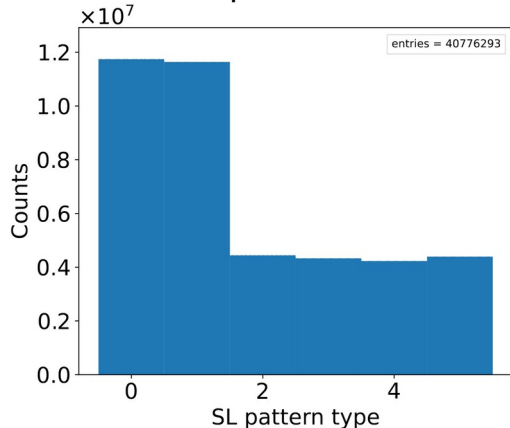
# DT superlayer patterns

- Perform triggerless track reconstruction:
  - Treat all cell hits as part of potential track
- Superlayer pattern:
  - Require track-like shape of hits
  - Hit time difference limited by max. drift time:
$$\Delta t_{\text{max}} \leq t_{\text{drift,max}}$$
  - Only accept patterns with 4 cell hits

Superlayer pattern geometry:



Pattern occupancies:



- Cumulated pattern rate: **700 Hz / superlayer**

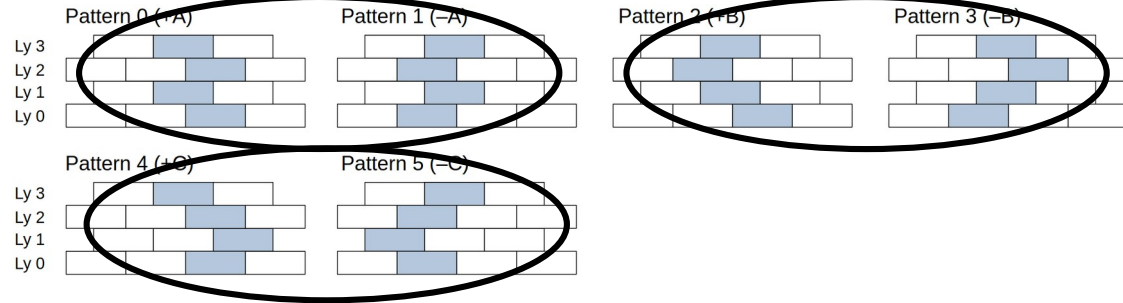
Atmospheric muon simulation: 450 Hz / superlayer

- Presence of noise-induced patterns in data

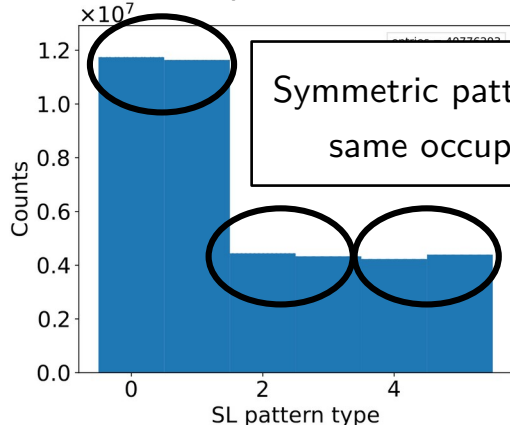
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Superlayer pattern geometry:



Pattern occupancies:



Simulated pattern rate: **700 Hz / superlayer**

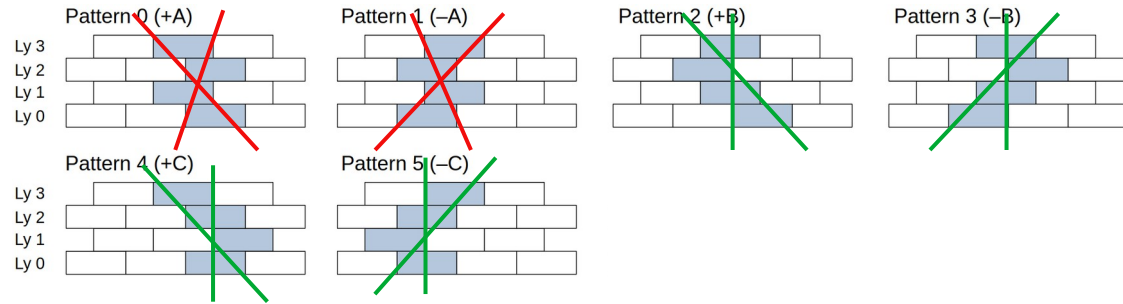
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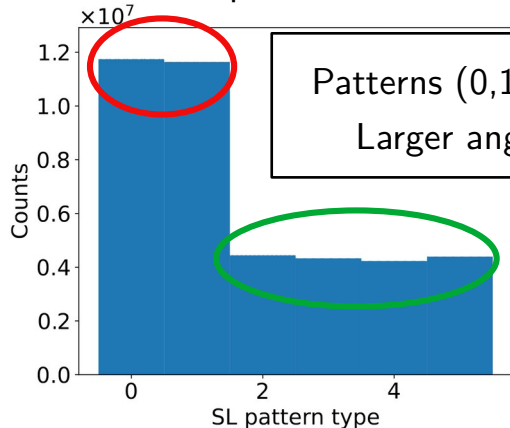
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$$\Delta t_{\max} \leq t_{\text{drift},\max}$$
  - Only accept patterns with 4 cell hits

Superlayer pattern geometry:



Pattern occupancies:



Patterns (0,1) have higher rate:  
Larger angular acceptance

pattern rate: **700 Hz / superlayer**

muon simulation: 450 Hz / superlayer

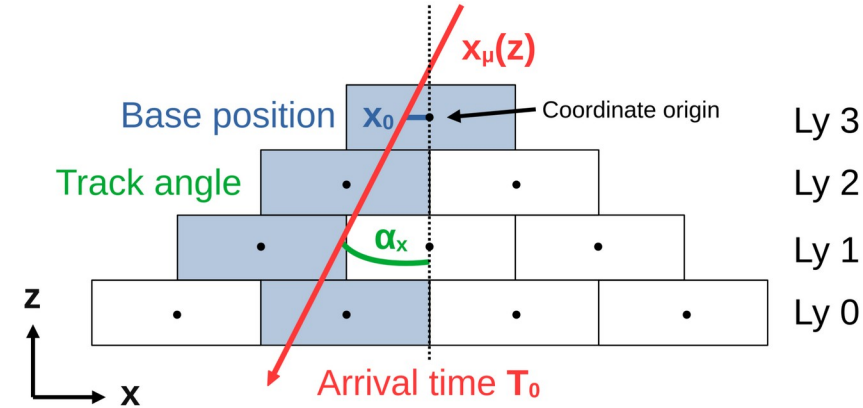
► Presence of noise-induced patterns in data

# DT superlayer pattern fit

- Assumptions for reconstruction:
  - Constant drift velocity over full length:  $\mathbf{v}_{\text{drift}} = \text{const.}$
  - Horizontal drift at height of wire:  $\mathbf{z}_{\text{hit},ly} = \mathbf{z}_{\text{wire},ly}$
  - Straight muon tracks:  $\mathbf{x}_{\text{hit},ly} = \mathbf{x}_0 + \mathbf{z}_{\text{wire},ly} \cdot \tan \alpha$
- Relation between cell hit position  $\mathbf{x}_{\text{hit},ly}$  and timestamp  $\mathbf{T}_{ly}$ :
  - Arrival time  $\mathbf{T}_0$  determines drift time:  $\mathbf{t}_{\text{drift},ly} = \mathbf{T}_{ly} - \mathbf{T}_0$
  - Drift distance:  $\mathbf{x}_{\text{drift},ly} = \mathbf{v}_{\text{drift}} \cdot \mathbf{t}_{\text{drift},ly}$
  - Left-right ambiguity of cell hit described by laterality  $\mathbf{L}_{ly}$ :  

$$\mathbf{x}_{\text{hit},ly} = \mathbf{L}_{ly} \cdot \mathbf{x}_{\text{drift},ly} + \mathbf{x}_{\text{wire},ly}$$
- Fit function of 4 timestamps ( $\mathbf{T}_{ly}$ ) to 3 track parameters ( $\mathbf{T}_0$ ,  $\mathbf{x}_0$ ,  $\tan \alpha$ ):

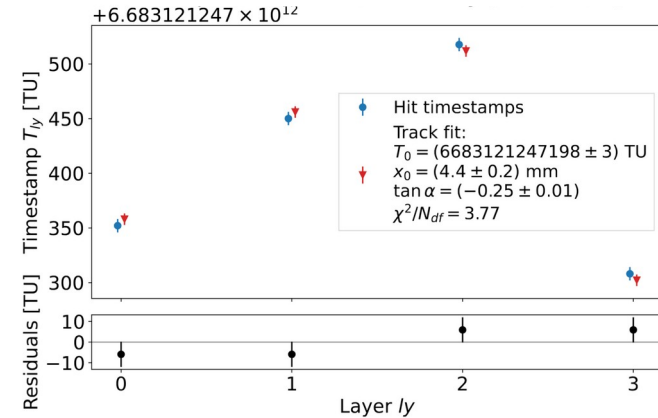
$$\mathbf{T}_{ly} = \mathbf{T}_0 + \mathbf{L}_{ly} / \mathbf{v}_{\text{drift}} \cdot ( \mathbf{x}_0 + \mathbf{z}_{\text{wire},ly} \cdot \tan \alpha - \mathbf{x}_{\text{wire},ly} )$$



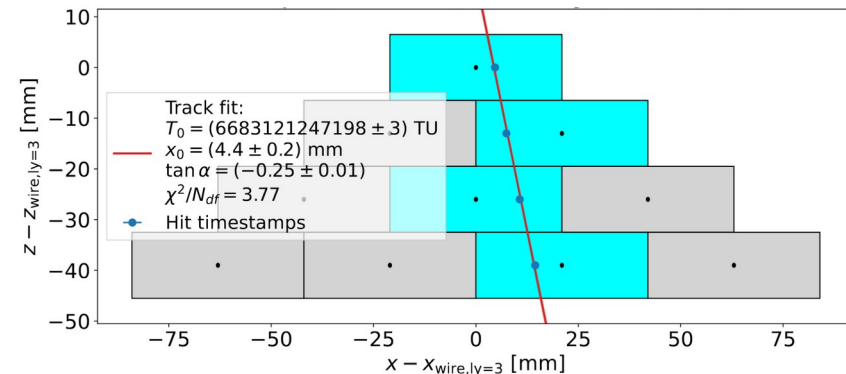
# DT superlayer pattern fit

- Perform timestamp fits for all individual patterns
  - Consider all laterality combinations and keep only best fit
- Apply cuts to fit results to suppress noise hits:
  - Goodness-of-fit variable:  $\chi^2/N_{df} < 10$
  - Unphysical drift times:  $2 \text{ ns} \leq t_{\text{drift}} \leq t_{\text{drift,max}} - 2 \text{ ns} = 383 \text{ ns}$
- ▶ Fitted pattern rate: **400 Hz / superlayer**
  - ▶ Atmospheric muon simulation: 450 Hz / superlayer
  - ▶ Relatively good agreement

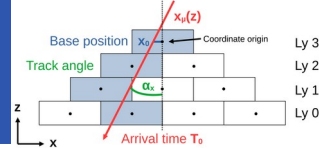
Example timestamp fit result:



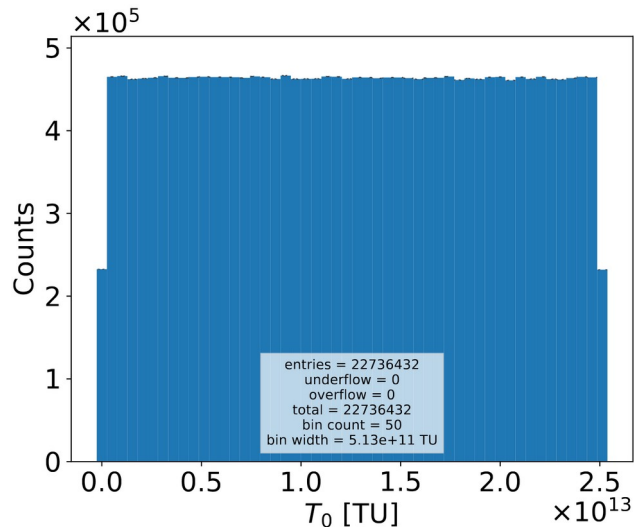
Reconstructed cell hit positions:



# DT superlayer pattern fit results

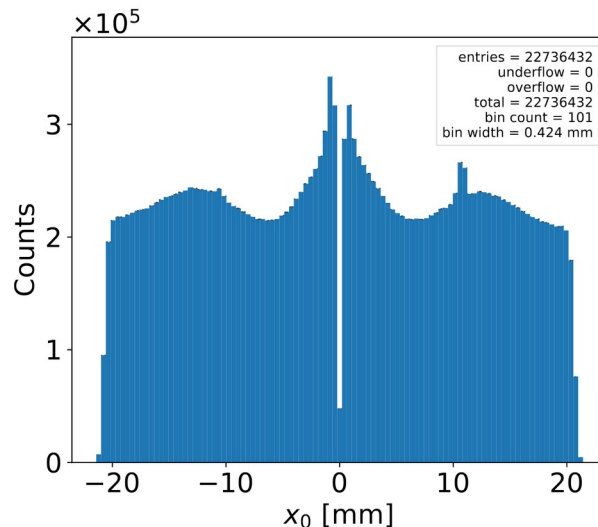


Arrival time  $T_0$ :



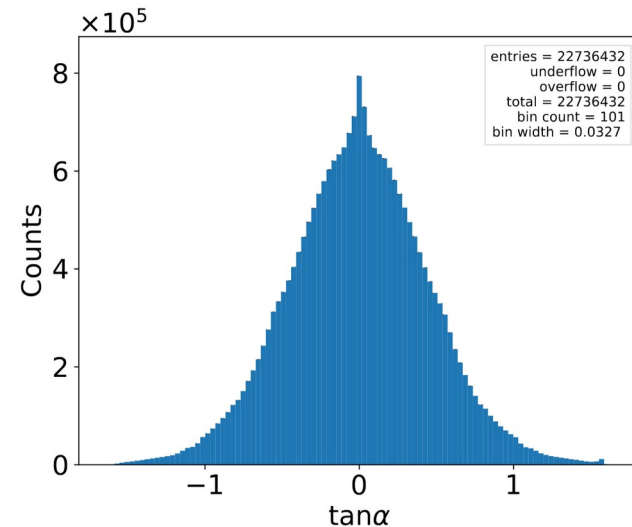
- Uniform distribution as expected for atmospheric muons

Reference position in cell  $x_0$ :



- Roughly uniform distribution in cell
  - Increased number of tracks close to wire: Field nonlinearity
  - Small feature at  $x_0=10.5$  mm: Noise hits?

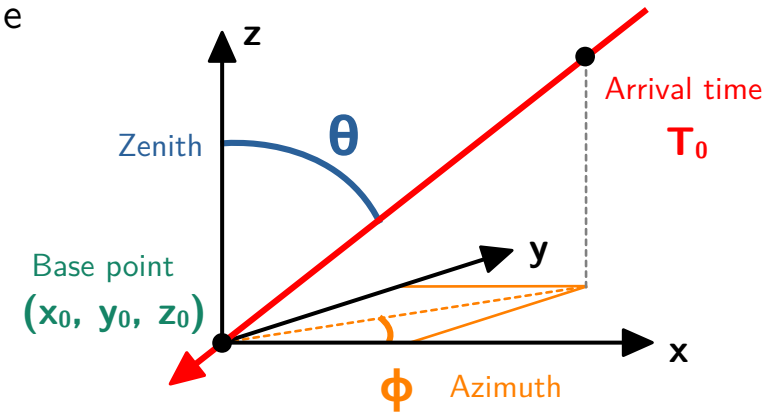
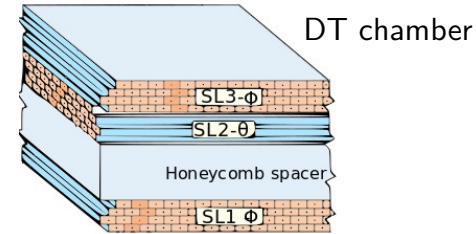
Track inclination  $\tan \alpha$ :



- Vertical tracks are preferred
- Symmetric shape
  - Small feature at  $\tan \alpha = 0$ : Noise hits?

# DT global track building

- Temporal matching of pattern fits in different superlayers
  - Select patterns close in time:  $\Delta T_{0,SL} \leq 20 \text{ ns}$
  - Remove pattern ambiguities
  - Check consistency of SLs 1, 3 pattern fits
- Combination of superlayer projections into global coordinate frame
  - Global track parameters:
    - Arrival time  $T_0$
    - Base point  $(x_0, y_0, z_0)$
    - Azimuth angle  $\Phi$ , zenith angle  $\Theta$
- ▶ Reconstructed global muon rate: **85 Hz**
- ▶ Global track parameters (mostly) behave as expected

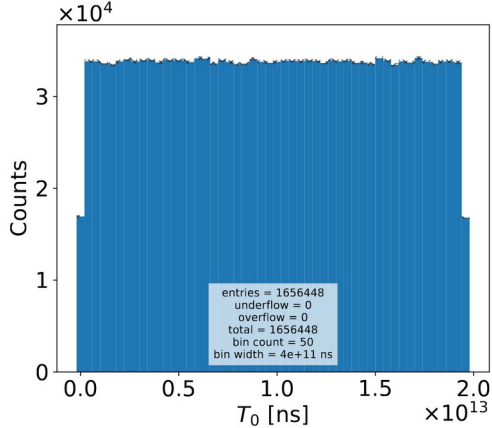


Atmospheric muon simulation: 196 Hz

- ▶ Lower rate of reconstructed muons: Possibly because of assumptions in reconstruction

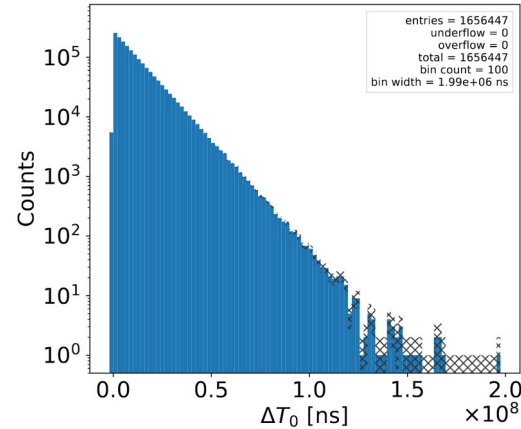
# DT global track results

Arrival time  $T_0$ :



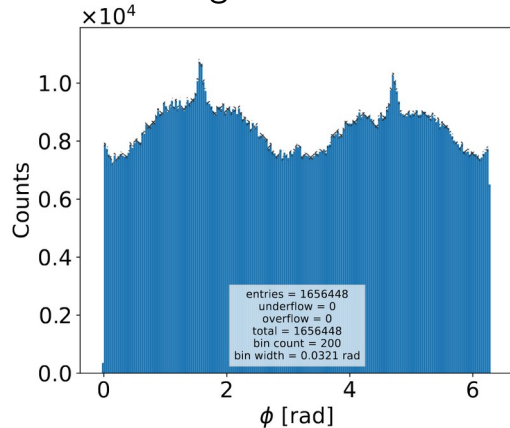
- ▶ Uniform distribution as expected for atmospheric muons

Time difference  $\Delta T_0$  of consecutive tracks:



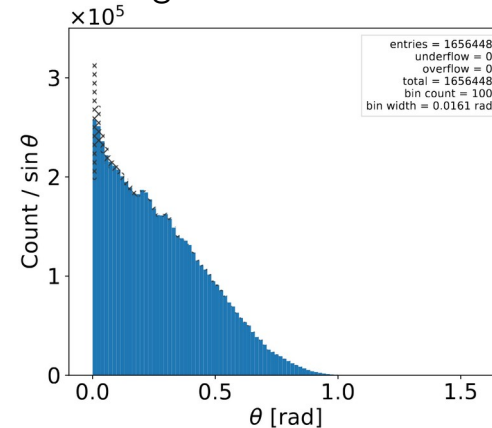
- ▶ Poisson-random arrival times of atmospheric muons leads to observed exponential distribution for time difference

Azimuth angle  $\Phi$ :



- ▶ Symmetry between  $[0, \pi]$  and  $[\pi, 2\pi]$  as expected
- ▶ Spike at  $\Phi = \pi/2, 3\pi/2$ : Noise hits?
- ▶ Non-uniform due to cuboid chamber shape

Zenith angle  $\Theta$ :



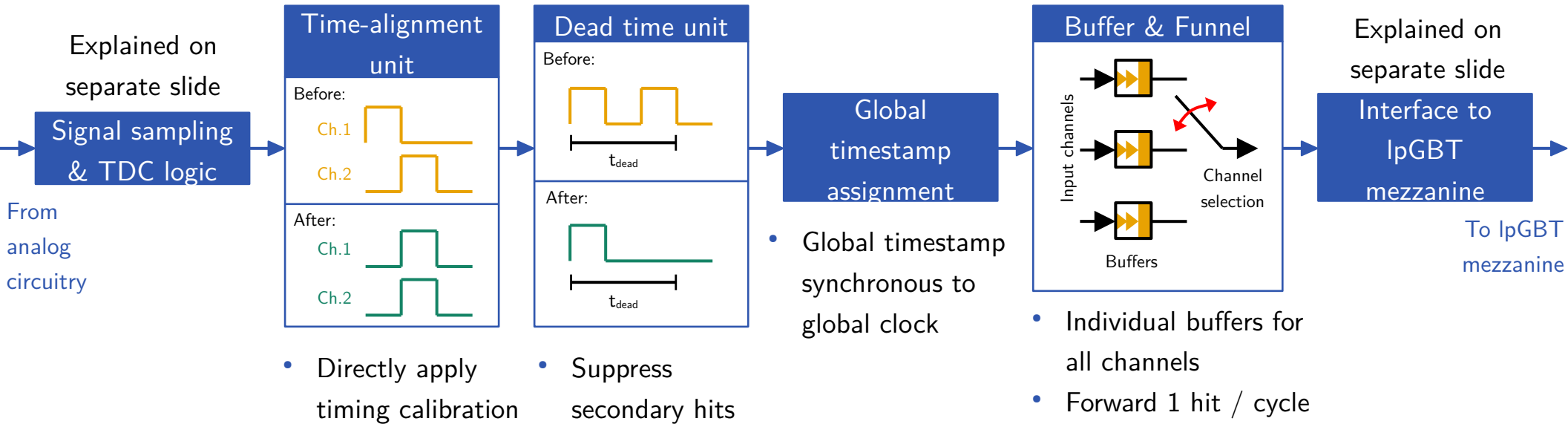
- ▶ Vertical tracks preferred
  - ▶ Matches qualitative expectation
- ▶ Steep decrease towards higher angles

## Backup (Pt. 3)

- SiPM-based scintillation detector
- Scintillator data analysis

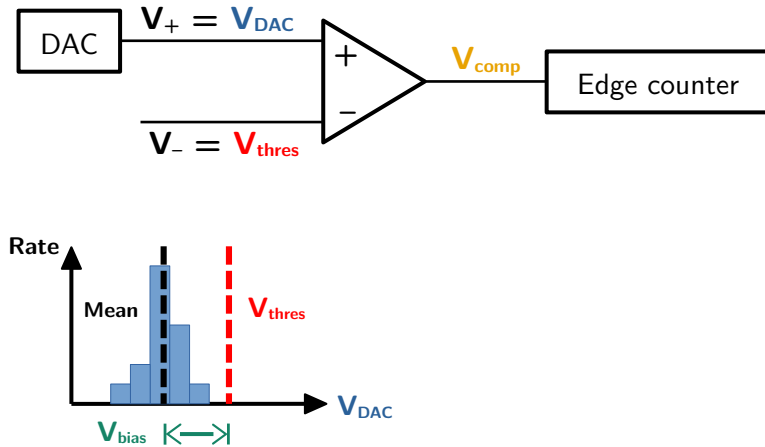
# Scintillator readout electronics – FPGA firmware

- Completely self-developed readout firmware running on FPGA:

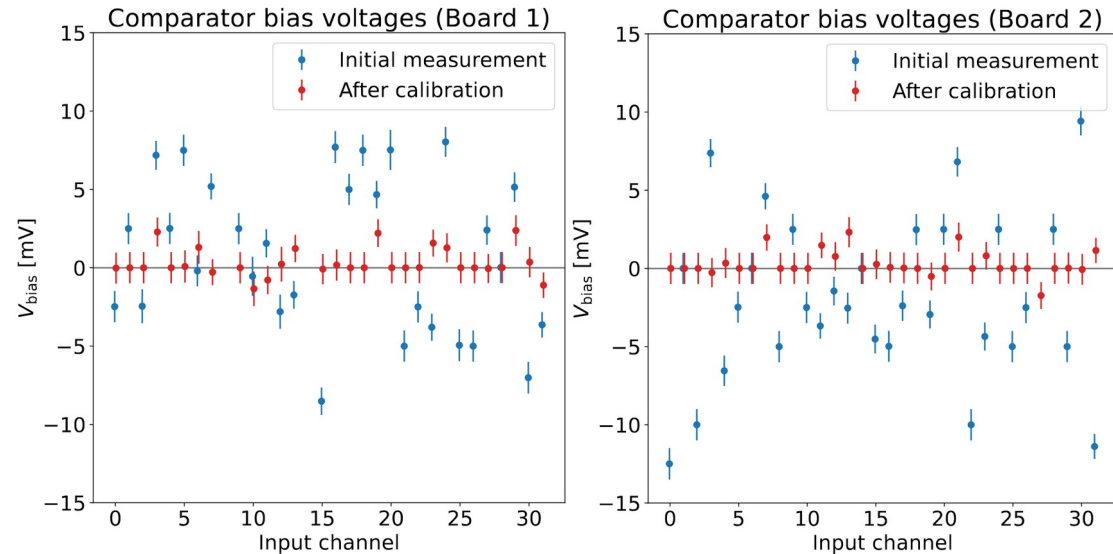


# Input discriminator threshold calibration

- FPGA-internal comparators are non-ideal
  - Switching point shifted by bias voltage  $V_{\text{bias}}$
- Bias measurement:
  - Scan comparator input voltage for fixed threshold
  - Determine comparator switching rate
  - Switching point lies at rate maximum

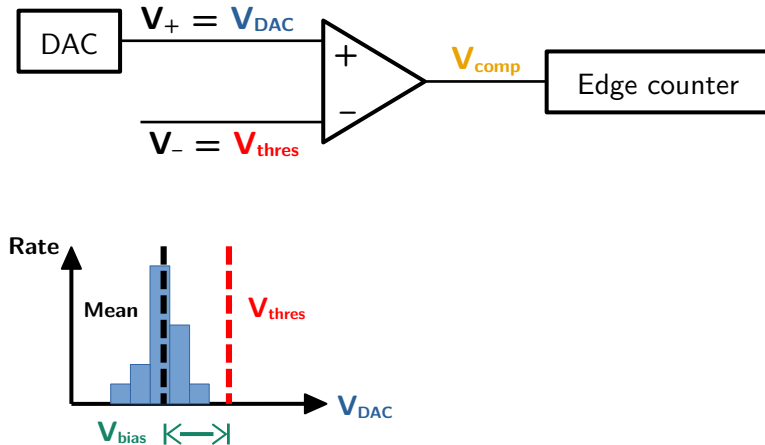


## Measured bias before and after correction:

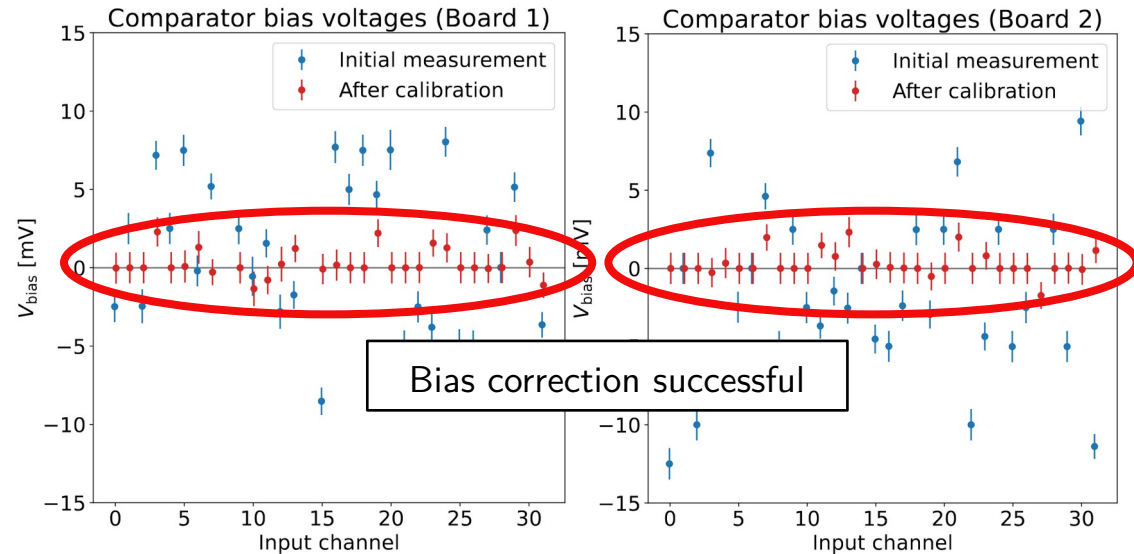


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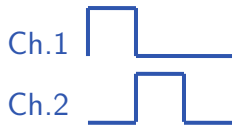
► Measured bias **before** and **after** correction:



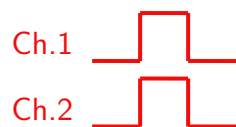
# Input timing calibration with testpulses

- TDC logic implemented in FPGA has non-deterministic timing offset
  - Due to chip-internal calibration of deserializers on startup
  - Affects groups of inputs (I/O banks)
- Timing offset measurement:
  - Send simultaneous testpulses to all inputs
  - Measure response timestamps
- Time-alignment unit:
  - Programmable delay units in FPGA firmware
  - Calculate target delays from testpulse timing

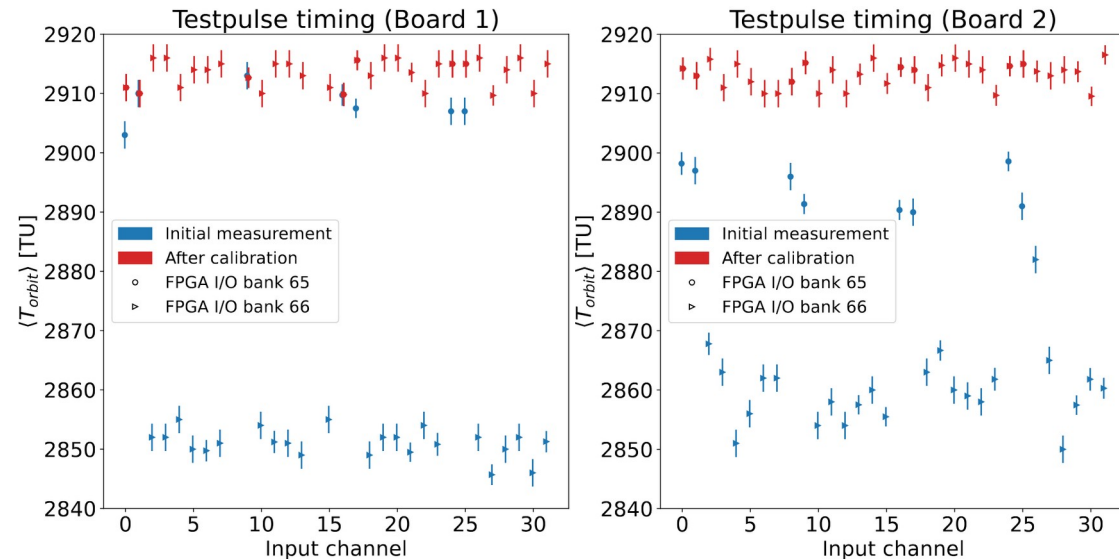
Before:



After:



- Measured testpulse response **before** and **after** programming delays:

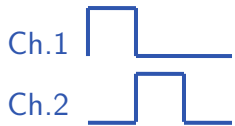


1 TU (timestamp unit) = 0.78 ns

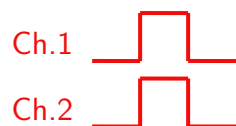
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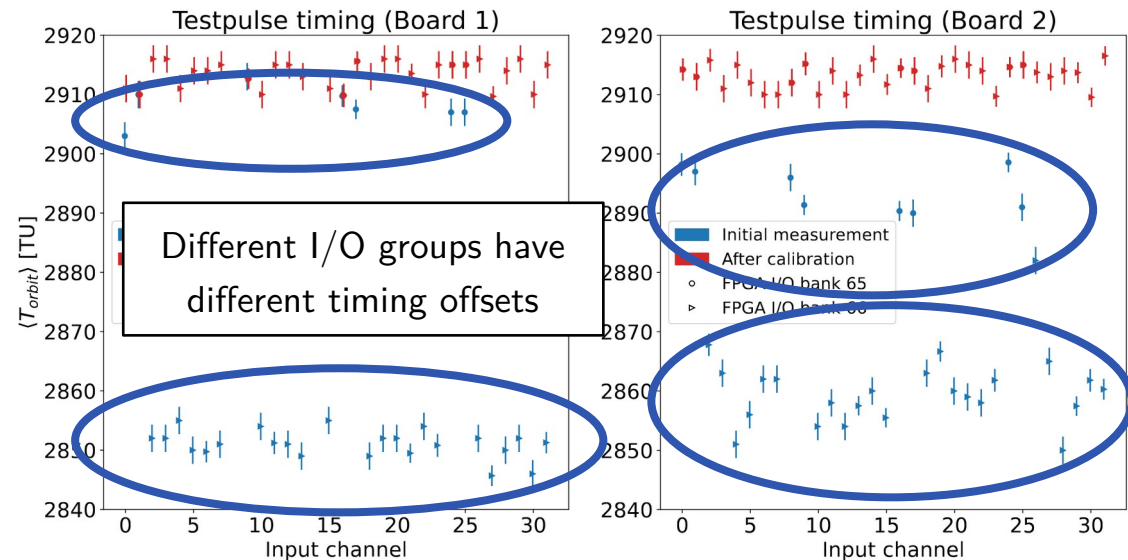
Before:



After:



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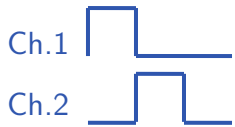


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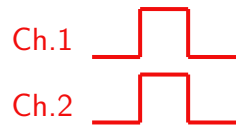
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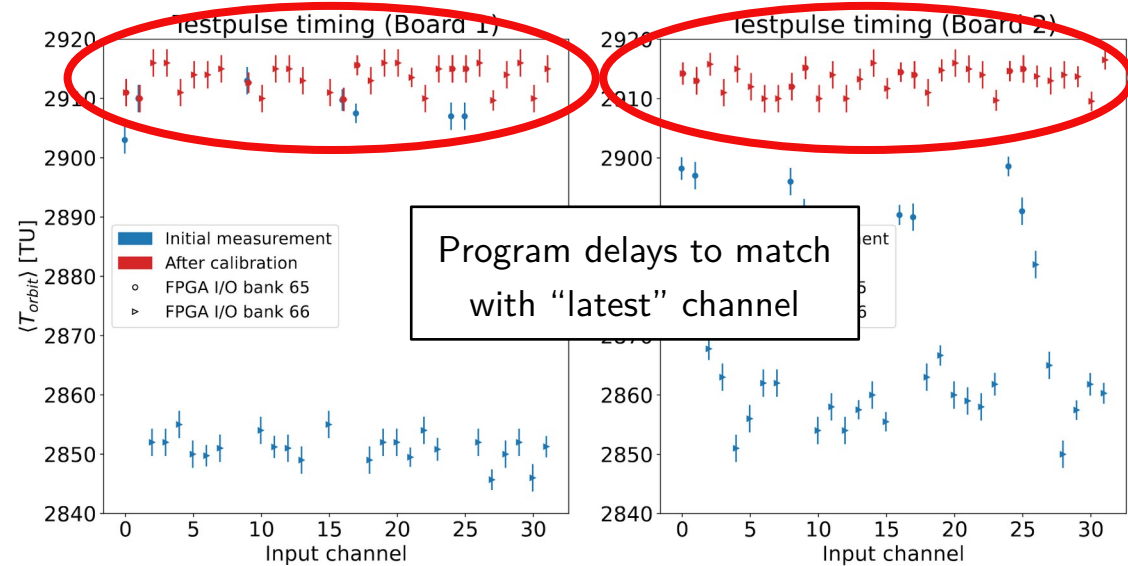
Before:



After:



- Measured testpulse response **before** and **after** programming delays:

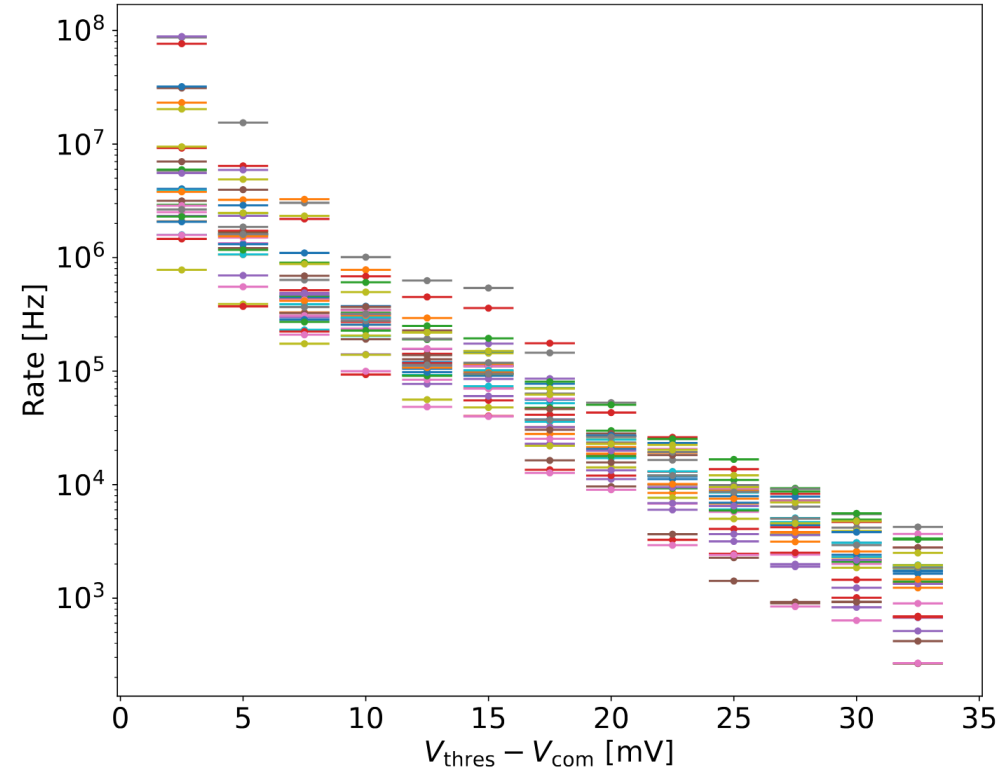


1 TU (timestamp unit) = 0.78 ns

# SiPM noise scan

- Performed measurement of SiPM noise spectrum
  - Scan threshold voltage range [2.5, 32.5] mV (or [0, 6] p.e.)
  - Measure SiPM rate with counter directly on FPGA
- ▶ **Exponentially decreasing rate**
  - ▶ Expected from noise & cell crosstalk:
    - ▶ Randomly triggered cell due to thermal noise: Hit with amplitude  $V_{\text{SiPM}} = 1$  p.e.
    - ▶ Crosstalk probability  $p$  with other cell allows higher amplitudes, however exponentially suppressed:  $P(V_{\text{SiPM}} = N \text{ p.e.}) = p^N$
    - ▶ Discrete steps in rate due to integer cell count
  - ▶ Expected steps in noise rate not visible
    - ▶ Aliasing effects as 1 p.e.  $\sim$  scan step size

Obtained rate spectrum for 29 SiPMs:

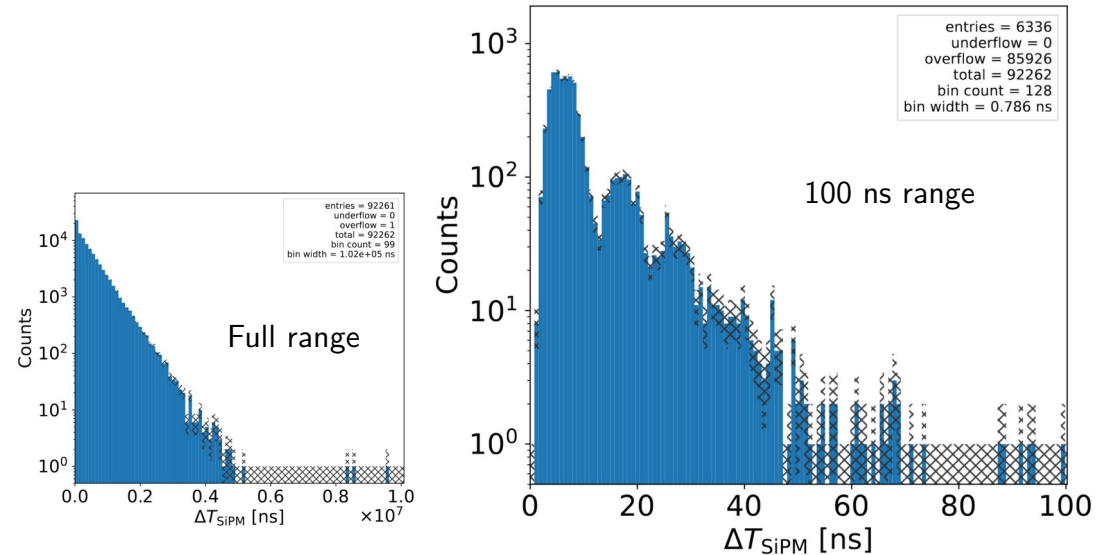


# SiPM hit data

## Without dead time:

- Observe systematic effects in SiPM hit data
  - High **SiPM rate**  $O(1 \text{ kHz})$  for selected threshold
  - **Secondary hits** with discrete timing
- ▶ Possible signal reflections or problems with sampling logic
  - ▶ Apply dead time as simple countermeasure

Time difference  $\Delta T_{\text{SiPM}}$  between hits of same SiPM:



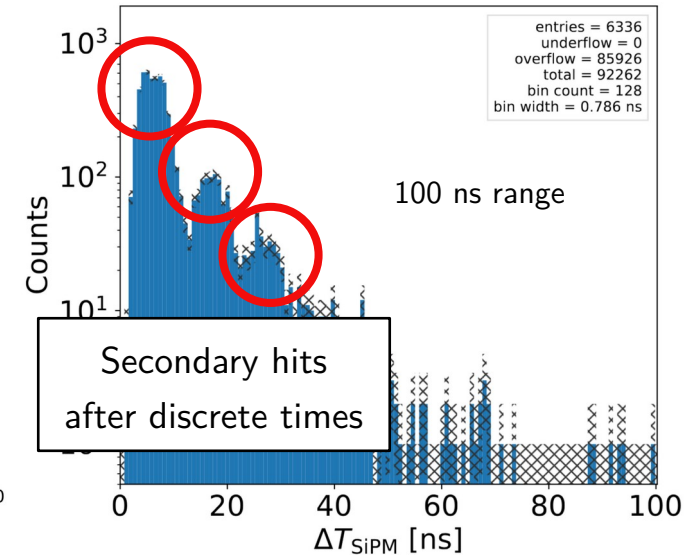
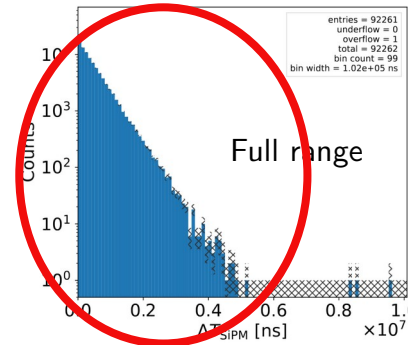
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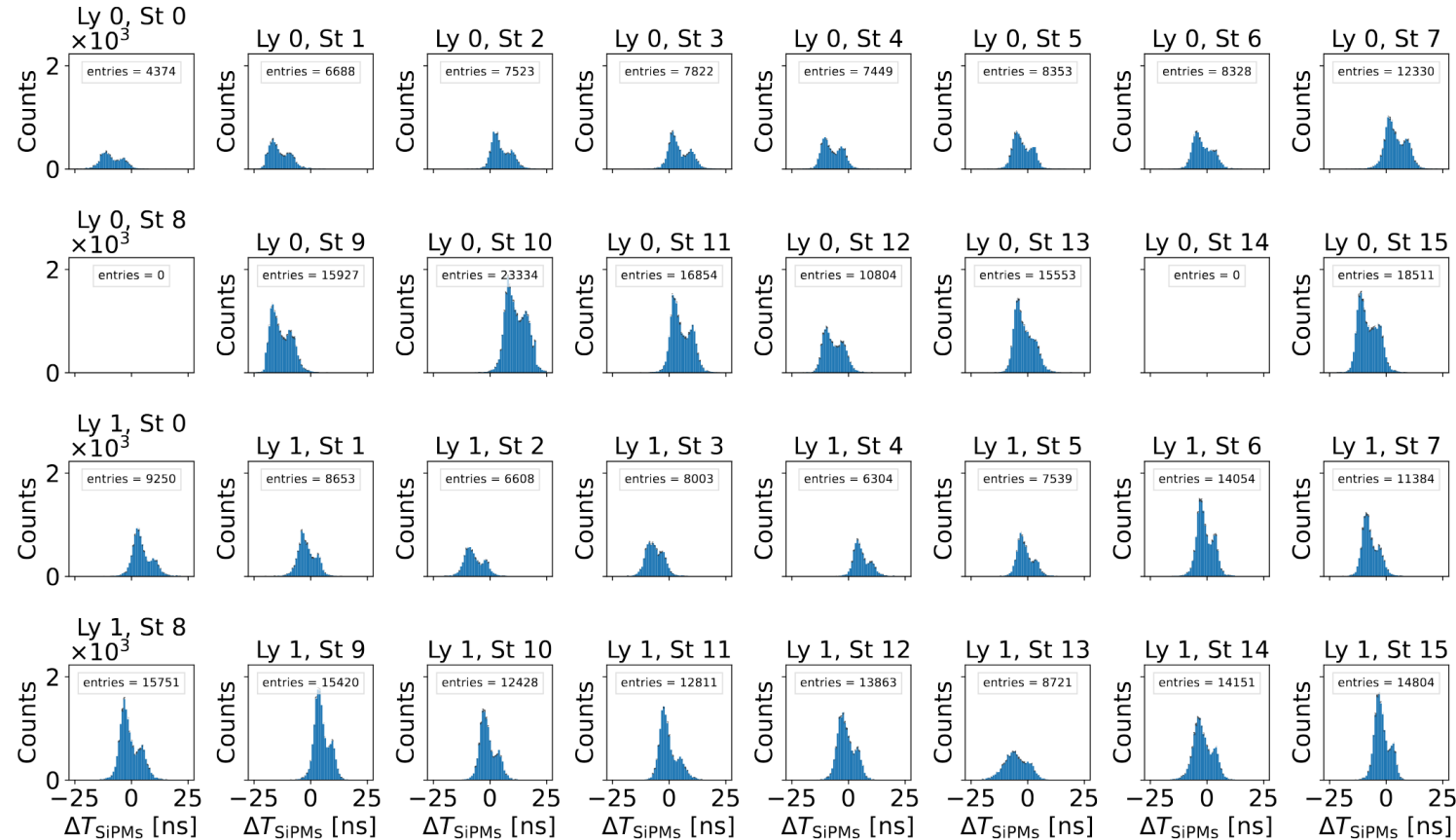
Exponential distribution:  
Poisson random events



# Scintillator strip coincidences

Time difference  $\Delta T_{\text{SiPMs}}$  of coincidences for all strips:

- Observe double-peak structure in time difference of SiPMs
- Needs future investigation
  - In this work, accept all coincidence hits, even though structure is visible



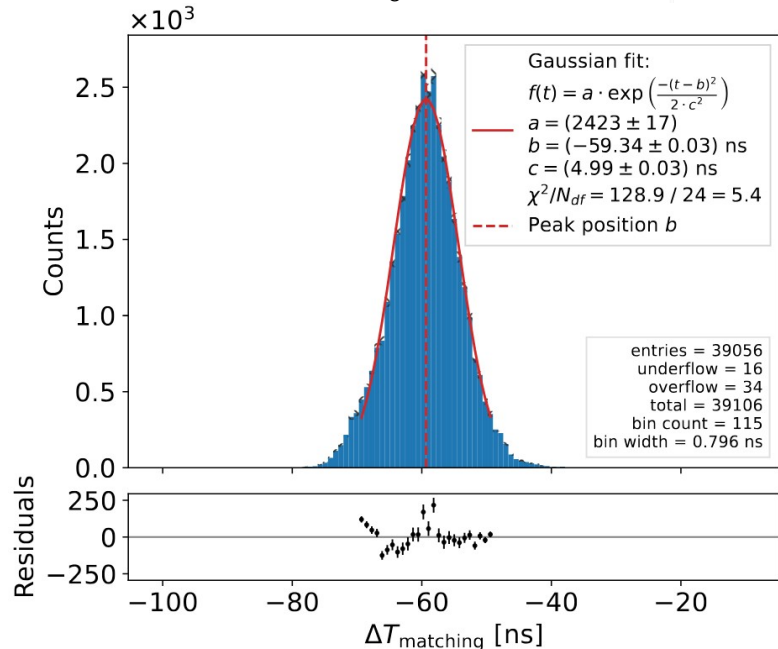
## Backup (Pt. 4)

- Combined data analysis

# Temporal matching of DT tracks & scintillator hits

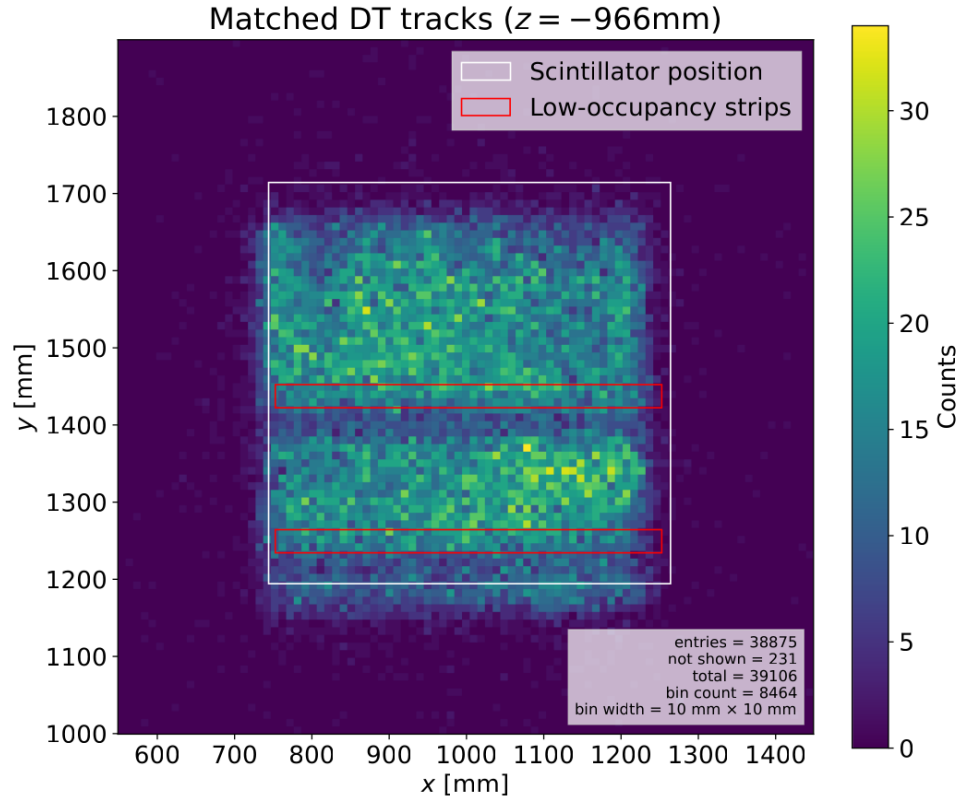
- Timing peak is not really Gaussian
  - Expected because SiPMs all have individual timing variations of up to 6 ns (uncertainty of scintillator timing calibration)
  - Rather: Sum of Gaussians of each individual scintillator strip

Time difference  $\Delta T_{\text{matching}}$  of matched hits with Gaussian fit:



# Independent cross-check of DT track reconstruction

2D histogram of matched tracks in scintillator:



- ▶ Fraction of matched DT tracks not hitting the scintillator is  **$O(1\%)$** 
  - ▶ See entries in stat-box in plot

## Backup (Pt. 5)

- Simplified Monte Carlo simulation with cosmics

# Monte-Carlo generation of atmospheric muon tracks

- Randomly generate atmospheric muon tracks
    - Base point (x,y): Chamber area  $\pm 1.5$  m, uniformly distributed
    - Base point z: Lowest point of chamber, fixed
    - Azimuth  $\Phi$ : Uniformly distributed
    - Zenith  $\Theta$ : Sample  $\sim \cos^2 \Theta$  distribution
  - Propagate tracks through detector geometry
    - DT chamber: Check for cell intersection at height of respective wire, Respect cell efficiency by “dicing”
    - Scintillator: Check for strip intersection at mean height
  - Perform exactly the same reconstruction methods as for data (discussed above)
- Very simplified “geometry only” approach !
- No modeling of noise & no modeling of spatial / timing uncertainty
  - No modeling of “physics”, i.e. muon interactions and effects in detector
  - Assumption of track hit at central height of detector device
    - At wire height in drift cell (strong simplification) & mean strip height in scintillator